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SRI SHANKARA MEENAKSHI SUNDARARAJAN RESEARCH CENTRE

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PROCEEDINGS OF FOURTH RESEARCH SUMMIT 2026

Explore, Evaluate, & Extend (RSEEE'26)

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Dr. Saravanan. S. V

Dr. Poongothai. S

Dr. Arulkarthick. V. J

Dr. Arivazhagan. R

Mr. Mohan Raj Vijayan

Dr. Sowmya.S

Proceedings of Fourth Research Summit 2026

Explore, Evaluate, & Extend (RSEEE'26)

Editors

Dr. Saravanan S.V
Dr. Poongothai S
Dr. Arul Karthick V. J
Dr. Arivazhagan R
Mr. Mohan Raj Vijayan
Dr. Sowmya S

Published by : KIM Scientific Publications, Chennai, India
Meenakshi Sundararajan Engineering College (Autonomous)
Arcot Road, Kodambakkam, Chennai - 600 024, Tamil Nadu, India
www.msec.edu.in
ISBN: **978-81-972100-0-6**
Meenakshi Sundararajan Engineering College (Autonomous), Chennai - 600 024
ISBN: 978-81-972100-0-6

About the Principal

Dr. Saravanan S.V is the Principal and Senior Professor with over 32 years of teaching, research, and administrative experience. He holds B.E., M.E., Ph.D. in Mechanical Engineering, M.B.A., and an Honorary D.Litt. degree, has guided seven Ph.D. scholars, mentored more than 100 university rank holders, holds published patent, and has authored textbooks with over 100 research publications. He serves as an editorial board member, has organized and chaired several national and international programs and conferences, and has received numerous prestigious academic and leadership awards.

About the Editors

Dr. Poongothai S is the Dean – Research & Welfare & Wellness (Learning Community) & former Professor & Head of Civil Engineering, Annamalai University with over 35 years of academic and research experience .She holds a Ph.D. from Anna University under QIP and M.E. degrees in Structures and Water Resources Engineering & Management, with extensive research output including International & National Journal & Conference publications more than 90, and a UGC-MRP funded project on groundwater quality modeling Co-Coordinator of One Crore UGC-SAP Project. She has guided 11 Ph.D. scholars and PG thesis, delivered numerous invited lectures, reviewer of reputed Journals and actively contributes to academic leadership, professional bodies, and research mentorship.

Dr. Arulkarthick V. J is Dean of the Internal Quality Assurance Cell and Professor in the Department of Electronics and Communication Engineering. He holds Ph.D. with a demonstrated history of working in the higher education industry. His research focuses on VLSI design, Image Processing and Communication Systems. His interest also includes Educational research, E-Learning, Faculty Development, Educational Leadership, and Classroom Management, Quality Assurance, Ranking & accreditation.

Dr. Arivazhagan R is Professor and Head of Research in the Department of Civil Engineering. He has 35 years of total experience, bringing extensive knowledge and expertise to academics and research. He holds an A.M.I.E. from The Institution of Engineers (India), an M.Tech in Coastal Management with University Second Rank from Anna University, and a Ph.D. in Geotechnical Engineering. He has made significant contributions to the field of Civil Engineering through teaching and research. He continues to guide and mentor future engineers with unwavering dedication.

ISBN: 978-81-972100-0-6

Mr. Mohan Raj Vijayan is an Assistant Professor and Research Member in the Department of Information Technology. He holds an M.E. in Computer Science and Engineering and is currently pursuing a Ph.D. in Knowledge-Based Systems at DIST, CEG, Anna University. His multidisciplinary qualifications span Management, Music, Psychology, Philosophy, French, and Archaeology & Epigraphy. With nearly two decades of experience in Academia, Research, and Industry, he brings strong practical and theoretical expertise, and his research interests include Generative AI, Intelligent Agents, Knowledge Engineering, Computational Linguistics, and Indian Knowledge Systems.

Dr. Sowmya S is an Associate Professor and Research Member holds a Ph.D. in Biomedical Signal Processing from Anna University, with research focused on wearable devices and deep learning models for maternal health monitoring having over 17 years of academic experience. Her contributions include publications in reputed SCI and Scopus-indexed journals, patents in healthcare technology, and active involvement as a reviewer for leading international journals. She is passionate about applying AI for real-time health diagnostics and has received awards for both academic excellence and innovative research.

About the Technical Editor

Dr. Ponni M holds a Ph.D. in Structural Engineering from Annamalai University, with her research specializing in passive and energy-efficient building systems. She brings over five years of academic experience complemented by seven years of industrial expertise, reflecting a strong blend of theoretical knowledge and practical application. Her scholarly contributions include publications in reputed Scopus-indexed journals, along with patents that demonstrate her commitment to innovation in the field of sustainable construction. She actively serves as a reviewer for leading international journals, contributing to the advancement of research quality and dissemination. She is deeply passionate about advancements in roofing systems, especially insulated and energy-efficient roof technologies.

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ISBN: 978-81-972100-0-6

VISION AND MISSION OF THE INSTITUTE

VISION

To impart state-of-the art technical education, including sterling values and shining character, producing engineers who contribute to nation building thereby achieving our ultimate objective of sustained development of an unparalleled society, nation and world at large.

MISSION

Meenakshi Sundararajan Engineering College, Chennai constantly strives to be a Centre of Excellence with the singular aim of producing students of outstanding academic excellence and sterling character to benefit the society, our nation and the world at large.

To achieve this, the college ensures

- Continuous upgradation of its teaching faculty to ensure a high standard of quality education and to meet the ever-changing needs of the society.
- Constant interaction with its stakeholders.
- Linkage with other educational institutions and industries at the national and international level for mutual benefit.
- Provision of research facilities and infrastructure in line with global trends.

Adequate opportunities and exposure to the students through suitable programs, to mould their character and to develop their personality with an emphasis on professional ethics and moral values.

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VISION AND MISSION OF THE RESEARCH CENTRE

VISION

To foster a vibrant research culture that promotes innovation, collaboration, and societal impact through interdisciplinary, industry-linked, and globally recognized research initiatives.

MISSION

- To foster a vibrant ecosystem of research excellence that empowers students and faculty in intellectual exploration and discovery.
- To advance knowledge and drive innovative solutions through impactful, publication-oriented research.
- To establish strong collaborations with national and international organizations, enhancing interdisciplinary engagement.
- To align institutional research initiatives with global Sustainable Development Goals (SDGs) and emerging technologies.
- To elevate the institution's research visibility, quality, and ranking through high-impact publications and citations.
- To contribute meaningfully to academia, industry, and the overall advancement of humankind.

ISBN: 978-81-972100-0-6

RESEARCH PILLARS

Proceedings of Fourth Research Summit 2026

Explore, Evaluate, & Extend (RSEEE'26)

30th April 2026

Meenakshi Sundararajan Engineering College (Autonomous), Chennai, Tamil Nadu – 600 024.

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Meenakshi Sundararajan Engineering College (Autonomous), Chennai - 600 024
ISBN: 978-81-972100-0-6

Book Title : Proceedings of Fourth Research Summit 2026
Explore, Evaluate, & Extend (RSISS'26)

Editor(s) Name : Dr. Saravanan S.V, Principal & Senior Professor
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Published By : KIM SCIENTIFIC PUBLICATIONS, CHENNAI, INDIA

Printer's Address : Meenakshi Sundararajan Engineering College (Autonomous)
Chennai – 600 024, Tamil Nadu, India

Edition Details : AI&DS/ Civil/ CSE/ ECE/ EEE/ IT/ MECH
CHE/ENG/PHY/MATHS/TAMIL

Month & Year : APRIL 2026

ISBN

9 788197 210006

Website

www.msec.edu.in

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ISBN: 978-81-972100-0-6

Dr. Saravanan S.V
Principal & Senior Professor

Message

"The best research will give the best teaching, and the best teaching will give the best research"

It gives me great pleasure to welcome you all to the Fourth Research Summit 2026 of our engineering college. This summit serves as a prestigious platform where innovative ideas, emerging technologies, and visionary research efforts come together.

In an era driven by knowledge, discovery, and rapid technological advancement, research is the cornerstone that shapes the future of engineering education. Our institution has consistently encouraged students and faculty to pursue research that is both socially relevant and academically rigorous. This summit reflects our continued commitment to nurturing a vibrant research culture on campus.

I extend my appreciation to all the presenters, participants, faculty mentors, and Dean- Research and organisers whose dedication has made this event possible and the proceedings with the ISBN. I am confident that the interactions, discussions, and collaborations emerging from this summit will inspire new perspectives and contribute meaningfully to the global scientific community.

Wishing you all a productive and enriching experience.

ISBN: 978-81-972100-0-6

Dr. Poongothai S
Dean -Research

Message

It gives me immense pleasure to extend my warm greetings to all participants of the Fourth Research Summit – RSEEE'26, hosted by the Sri Shankara Meenakshi Sundararajan Research Centre (SSMSRC) at Meenakshi Sundararajan Engineering College. This summit stands as a distinguished platform celebrating research excellence, collaborative innovation, and transformative engineering solutions.

The theme of this year is "Explore, Evaluate, & Extend " captures the essence of modern research and the spirit of technological advancement shaping the world today. Innovation fuels new ideas, simulation empowers experimentation and validation, and solution-driven engineering transforms concepts into impactful realities. Together, these principles form the foundation of progressive research that addresses societal and industrial challenges. This summit brings together brilliant minds researchers, academicians, industry experts, and students to exchange knowledge, explore emerging technologies, and engage in purposeful discussions.

I sincerely appreciate the relentless efforts of the authors, reviewers, session chairs, coordinators, and the organizing committee whose dedication has shaped this remarkable academic gathering. My heartfelt congratulations to all contributors for enriching our research culture with high-quality scholarly work. Let this summit be an inspiration to think beyond boundaries, simulate without constraints, and solve with purpose, advancing research that benefits society and humanity. I wish RSISS'26 grand successes and encourage every participant to seize this opportunity to collaborate, innovate, and lead the future of engineering.

I profusely thank our beloved Secretary, Principal, Heads of the department and Management for their consistent support to bring this Research Summit a phenomenal success.

Dr. Ramajeyam L
Dean -PG Studies

Message

The Research Summit at Meenakshi Sundararajan Engineering College reflects our institution's enduring commitment to academic excellence, innovation, and societal progress. This year's collection of research contributions highlights the intellectual curiosity, technical expertise, and collaborative spirit that define our students and faculty. From emerging engineering technologies to impactful interdisciplinary studies, each project embodies our vision of fostering knowledge that addresses real-world challenges and promotes sustainable advancement.

At MSEC, we believe that research is not merely an academic endeavor but a catalyst for transformation. The works showcased in this summit illustrate the power of inquiry-driven learning, creative thinking, and persistent exploration. I extend my appreciation to all researchers, mentors, and organizers whose dedication has elevated the standard of scholarship within our college.

I invite every participant to engage deeply with the ideas presented, exchange perspectives, and forge new pathways for innovation. May this summit inspire continued excellence and strengthen our collective pursuit of engineering solutions that benefit society.

ISBN: 978-81-972100-0-6

Dr. Arul Karthick V.J
Dean -IQAC

Message

It gives me great pleasure to present the Proceedings of the Research Contributions submitted by scholars, faculty members, and students. This compilation reflects the institution's continuous commitment toward academic excellence, innovation, and quality enhancement.

The Internal Quality Assurance Cell (IQAC) consistently strives to promote a culture of research and to ensure that scholarly activities align with our vision of achieving national and global standards. These proceedings stand as a testament to the dedication, hard work, and intellectual strength of our academic community.

I appreciate the efforts of all contributors, coordinators, reviewers, and organizing members who have worked diligently to bring this volume to fruition. I wish all researchers continued success in their academic pursuits and encourage them to carry forward the spirit of inquiry and innovation.

Dr. Arivazhagan R
Head - Research

Message

This Research Summit provides a premier multidisciplinary platform for researchers, scholars, educators, and students to present and discuss the latest innovations, emerging trends, and practical challenges in the fields of Engineering and Technology. It also highlights the solutions and advancements adopted to address these challenges. The Summit serves as a bridge connecting students with industry, exposing them to real- world problems, high-level innovations, advanced laboratories, and meaningful research opportunities that enable them to work on real-time solutions.

This Summit showcases the significant progress our institution has made in strengthening and enhancing our research and publication capabilities. It also inspires us to pursue impactful innovations that will contribute to the betterment of society and mankind. We would like to express our sincere gratitude to all the authors and reviewers whose contributions have enabled us to maintain the high quality of the manuscripts published in the SSMSRC proceedings.

We remain optimistic that the proceedings of Research Summit 2026 will, in due course, be indexed in IEEE Xplore, Web of Science, and Scopus. We appreciate that the authors of Research Summit 2026 may wish to enhance the visibility and reach of their work, and we assure them of our full support in these efforts. We trust that the findings and recommendations emerging from this Summit will serve as a strong impetus for further research and scholarly exploration in these areas. We extend our heartfelt thanks to all authors and participants for their valuable contributions.

ISBN: 978-81-972100-0-6

Prof. Raman S
Department of Mathematics

Message

It gives me immense happiness to see our Fourth Research Summit emerge as a collective expression of curiosity, responsibility, and humaneness. For twenty-five years, our institution has grown on a foundation of values, and this summit reflects that heritage.

The research presented here is not merely academic each contribution seeks to address real challenges, offer sustainable solutions, and serve society with integrity. I deeply appreciate the dedication of our faculty, whose work continues to inspire our students and uphold the ethical spirit that defines our college. May this pursuit of knowledge guide us toward a brighter, more compassionate future.

ISBN: 978-81-972100-0-6

PREFACE

It is with great pleasure that we present the Proceedings of the Fourth Research Summit 2026 to be held on April 30, 2026 at Meenakshi Sundararajan Engineering College. This annual summit serves as a significant platform for promoting scholarly exchange, showcasing innovative research, and encouraging academic engagement among students, faculty members, and researchers.

The Fourth Research Summit reflects the institution's continued commitment to advancing research-driven learning and fostering a culture of inquiry across diverse engineering and interdisciplinary domains. The abstracts included in this volume have undergone a rigorous review process to ensure academic quality and relevance. The breadth of topics covered demonstrates the expanding scope of research activities carried out within the college and in collaboration with the wider academic community. This proceeding aims to document the intellectual contributions presented at the summit and to inspire future investigations, collaborations, and innovations. It is our hope that the research compiled here will contribute meaningfully to ongoing academic discourse and provide a foundation for continued advancements in engineering and technology.

The Editorial Committee extends its sincere appreciation to all contributors whose efforts have enriched the Proceedings of the Fourth Research Summit 2026 of Meenakshi Sundararajan Engineering College. We thank all authors for their scholarly submissions and the reviewers for their meticulous evaluations and constructive feedback, which have helped uphold the academic quality of this publication. We also acknowledge the invaluable support of the session chairs, faculty coordinators, technical staff, and student volunteers whose dedication has ensured the smooth conduct of the summit. Our heartfelt gratitude goes to the Management of Meenakshi Sundararajan Engineering College for their unwavering encouragement and for providing the necessary resources to facilitate this academic endeavour.

The Research Summit provides an opportunity to explore emerging ideas, share innovative findings, and engage in meaningful discussions across various domains of engineering and technology. We hope that today's deliberations will spark new perspectives, inspire collaborative endeavours, and contribute to the continued advancement of research within our institution.

We extend our best wishes to all participants for a productive and enriching experience.

-Editorial Committee
Fourth Research Summit 2026
Meenakshi Sundararajan Engineering College

ISBN: 978-81-972100-0-6

Department of Artificial Intelligence and Data Science

Mrs. Mathangi N
Head

Message

As Head of the Department, I am proud to witness our faculty and students consistently pushing the boundaries of knowledge through impactful research. Our department fosters a culture of curiosity, collaboration, and innovation that translates into meaningful real-world solutions.

Each research contribution reflects our commitment to academic excellence and societal advancement. We continue to encourage interdisciplinary exploration and nurture emerging scholars. Together, we aim to build a vibrant research ecosystem that inspires progress and discovery.

ISBN: 978-81-972100-0-6

Department of Civil Engineering

Dr. Asha P
PG Head

Message

It gives me great pleasure to extend my warm greetings to all participants of this year's Research Summit. This summit stands as a platform where ideas converge, perspectives deepen, and innovative thinking takes flight. As we witness rapid advancements across disciplines, the role of research has never been more vital. It is through rigorous inquiry and collaborative exploration that we shape solutions for the challenges of tomorrow.

Our institution has always encouraged a culture of curiosity, critical thinking, and multidisciplinary engagement. I am proud of the dedication shown by our faculty, scholars, and students who continuously strive to push the boundaries of knowledge. Events like this reflect our shared commitment to academic excellence and meaningful societal impact. I encourage all participants to engage wholeheartedly—ask bold questions, share your insights generously, and forge collaborations that extend beyond this summit. Let this gathering inspire new ideas, strengthen research ecosystems, and open pathways for impactful innovation.

My sincere appreciation goes to the organizers, contributors, and distinguished speakers for making this summit possible. I wish everyone an enriching and inspiring experience.

ISBN: 978-81-972100-0-6

Department of Civil Engineering

Mrs. Nirmalambal U

Head

Message

It is a great honor and privilege to extend my warm greetings to all participants of the Fourth Research Summit 2026. This summit serves as an inspiring platform that brings together researchers, academicians, industry experts, and students to share innovative ideas, exchange knowledge, and foster collaborative opportunities.

In today's rapidly evolving world, research plays a vital role in addressing global challenges and driving sustainable development. Events like this summit not only encourage intellectual discussions but also nurture creativity, critical thinking, and interdisciplinary approaches.

My heartfelt appreciation goes to the organizers for their dedication in bringing together this remarkable platform for academic growth and innovation. I also acknowledge the participants for their insightful contributions that will strengthen the research community.

I wish the summit great success and hope it leads to insightful discussions, impactful outcomes, and lasting collaborations.

ISBN: 978-81-972100-0-6

Department of Computer Science and Engineering

Dr. Arthi S
Head

Message

It is my pleasure to extend warm greetings to all participants of the Research Summit 2026. This forum provides an excellent opportunity for sharing innovative ideas, discussing emerging trends, and encouraging collaborative research in Computer Science and Engineering.

Our department remains committed to fostering a strong research culture and empowering students and faculty to contribute meaningfully to the rapidly evolving technological landscape.

I appreciate the efforts of the organizing committee, reviewers, and contributors who have worked diligently to make this event impactful. I also congratulate all authors for their scholarly contributions. May this summit ignite new collaborations, inspire fresh ideas, and strengthen our collective commitment to academic excellence and societal advancement.

I wish the Research Summit great success and look forward to seeing the innovations and discussions that will emerge from this gathering.

ISBN: 978-81-972100-0-6

Department of Electronics and Communication Engineering

Mrs. Siji Sivanandhan
Head

Message

The summit aims to bring together experts, researchers, and students to share their knowledge and ideas on cutting-edge technologies. The summit will feature paper presentations on topics including Artificial Intelligence, IoT, VLSI, and Renewable Energy. Selected papers will be published in our college proceedings. Looking forward to your participation.

ISBN: 978-81-972100-0-6

Department of Electrical and Electronics Engineering

Dr. Maya R
Head

Message

It is my great pleasure to extend greetings to all attendees of the Research Summit 2026. This Summit will be an excellent opportunity, will contribute to the advancement of research, knowledge gathering, and collective effort in the Field of AI.

The Department of Electrical and Electronics Engineering is committed to the Technological Advancements, Impactful Innovations, Addressing the challenges in the Society. Artificial Intelligence has Impact on all sectors of society, this conference will contribute to its advancement and advanced research.

My heartfelt congratulations to the all the organizers, authors, editors and reviewers. I wish the research summit will achieve the great success. Wishing everyone enriching and enlightening experience.

ISBN: 978-81-972100-0-6

Department of Information Technology

Dr. Kanmozhi A
Head

Message

It is my pleasure to extend warm greetings to all participants of the Research Summit 2026. This gathering offers a meaningful opportunity for the exchange of knowledge, exploration of industrial advancements, and the encouragement of collaborative endeavors in Information Technology.

Our department remains committed to cultivating a dynamic research culture and enabling both students and faculty to contribute significantly to the fast- changing technological landscape. I sincerely commend the dedication of the organizing team, reviewers, and all contributors whose collective efforts have made this event truly impactful. My heartfelt congratulations also go to the authors for their noteworthy academic contributions.

May the Research Summit 2026 achieve outstanding success, inspire innovative thinking, spark fruitful partnerships, and reinforce our shared pursuit of academic excellence and societal growth.

ISBN: 978-81-972100-0-6

Department of Mechanical Engineering

Dr. Thiyagu M
Head

Message

The Department of Mechanical Engineering proudly announces its participation in the Fourth Research Summit organized by the Sri Shankara Meenakshi Sundararajan Research Centre of Meenakshi Sundararajan Engineering College. Scheduled on 30th April 2026 at 8:30 A.M. in KRS Hall, this summit serves as a significant academic platform to promote research culture, interdisciplinary collaboration, and innovation among faculty members, researchers, and students. As the Head of the Department, I view this summit as a vital initiative that aligns with our department's vision of fostering research-driven learning and advancing technological competence. Mechanical Engineering, being a core discipline, plays a crucial role in addressing real-world challenges through design innovation, materials development, thermal systems, manufacturing advancements, and sustainable engineering solutions. This summit provides an excellent opportunity for our faculty and students to showcase their research contributions, exchange ideas, and engage with experts from diverse domains. The Fourth Research Summit brings together academicians, industry professionals, faculty members and research scholars to discuss emerging trends and innovations in engineering. It promotes the presentation of research papers and deliberations on areas such as advanced manufacturing, computational analysis, energy systems, smart materials, and Industry 4.0, thereby bridging the gap between theory and industrial practice. The Department of Mechanical Engineering actively focuses on research in composite materials, thermal engineering, and simulation-based design. Participation in this summit will strengthen our research ecosystem through collaborations, improved publication quality, and enhanced funding opportunities, while also encouraging students to develop inquiry-based learning and problem-solving skills.

ISBN: 978-81-972100-0-6

Department of Sciences and Humanities

Mrs Vanathi T
Head

Message

I am very much excited that our 'Sri Shankara Meenakshi Sundararajan Research Centre' is hosting the *4th Research Summit* on 30th April 2026. I am sure this summit would offer an excellent platform to showcase our work, exchange ideas across disciplines, and strengthen our research community. Looking forward to a vibrant and productive summit. All the very best!

ISBN: 978-81-972100-0-6

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ISBN: 978-81-972100-0-6

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ISBN: 978-81-972100-0-6

MSEC-RS-2026-PR-001

Comparative Structural and Fatigue Analysis of ATV Steering Knuckle Using Aluminium 7050-T7451 Over Aluminium 6061-T6

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Abstract

Steering knuckle is considered one of the most critical load-bearing components in an automotive suspension system. It is subjected to complex, time-varying multi-axial forces during service life — including cornering, braking, and road-induced impacts — leading to fatigue failure. This study presents a comprehensive comparative finite element analysis (FEA) of an ATV steering knuckle manufactured from two aluminium alloys: Aluminium 6061-T6 (existing material) and Aluminium 7050-T7451 (proposed material). A geometrically accurate three-dimensional model was developed in SolidWorks based on Baja SAE competition vehicle parameters and analysed in ANSYS 2025 R2 using a 20-node hexahedral Hex20 mesh (495,282 nodes; 108,177 elements). Boundary conditions — cornering force (6,400 N), bump force (6,200 N), and braking force (2,400 N) — were analytically derived from vehicle dynamics. Results reveal identical von-Mises stress distributions (77.16 MPa) for both materials, while Al 7050-T7451 achieves a minimum Factor of Safety of 2.07 against 1.56 for Al 6061-T6, representing a 32.91% structural safety improvement. Total deformation is reduced by 3.8%. These findings confirm Al 7050-T7451 as the superior material for ATV steering knuckle applications demanding high fatigue resistance, stiffness, and structural safety.

Keywords: *Steering Knuckle, Finite Element Analysis, Al 7050-T7451, Al 6061-T6, Fatigue Analysis, Factor of Safety, ANSYS Simulation.*

MSEC-RS-2026-DR-002

A Scalable Explainable Geo-AI Framework for Spatio-Temporal Groundwater Quality Prediction under Climate Variability

Poongothai S[1], Dean-Research and Professor, Mohan Raj Vijayan[2], Assistant Professor
[1] Department of Civil Engineering, and [2] Department of Information Technology
[1, 2] Meenakshi Sundararajan Engineering College, Chennai 600 024, Tamil Nadu, India.
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Abstract

Evaluation of groundwater quality is very important for the sustainable utilization of groundwater resources, especially in semi-arid zones that face escalating environmental stress conditions. Despite this, traditional methods like WQI (Water Quality Index) and machine learning alone have failed to account for the dynamic nature of the processes involved in groundwater quality changes and have proven to be relatively static, local, and backward-looking methods, thus hindering their application in prediction of future trends of contamination. In response to these issues, this study presents an intelligent, scalable approach for predicting spatiotemporal groundwater quality by employing deep learning, spatial analysis through GeoAI techniques, and physics-based modeling approaches. This AI based modeling considers different sources of geospatial information, hydro chemical information, and climate forecasting to model the changes in the water quality of groundwater at spatial and temporal dimensions. Hybrid modeling based on the integration of spatiotemporal neural networks model and transfer learning approach is considered in this study to increase the accuracy rate and enable the use of this model for multiple regions under diverse hydrogeologic settings. Furthermore, explainable AI approaches are applied in this study to understand the causes behind water quality changes.

Key words: *Groundwater Quality Prediction; Spatio-Temporal Modeling; Explainable AI (XAI); Geo-AI; Climate Projection Integration*

MSEC-RS-2026-DP-003

Structural Health Monitoring of Buildings Using Artificial Intelligence for Early Damage Detection

Ramajeyam L, Dean PG Studies and Professor,
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Abstract

Structural Health Monitoring (SHM) is essential for ensuring the safety, durability, and performance of building structures, particularly under aging, environmental effects, and seismic loads. Conventional inspection techniques are often labour-intensive, time-consuming, and incapable of identifying early-stage damage. This study presents an Artificial Intelligence (AI)-based framework for early damage detection in building structures using structural response data. A numerical model of a multi-storey reinforced concrete building is developed, and damage scenarios are simulated by introducing stiffness degradation in selected structural elements. Dynamic response parameters, including natural frequencies and mode shapes, are extracted and used as input features for machine learning models. Supervised learning algorithms such as Artificial Neural Networks (ANN) and Support Vector Machines (SVM) are trained to identify and classify damage conditions. The performance of the proposed approach is evaluated using accuracy, precision, and recall metrics. The results demonstrate that the AI-based model can effectively detect early-stage damage with high reliability, significantly improving the efficiency of structural assessment. The study highlights the potential of integrating artificial intelligence with structural engineering practices for real-time monitoring and predictive maintenance of buildings.

Keywords: *Structural Health Monitoring, Early Damage Identification, Vibration-Based Monitoring, Predictive Maintenance Smart Infrastructure.*

MSEC-RS-2026-DI-004

Deep Learning-Based Multimodal Image Fusion using CNN and GAN for Improved Visual Fidelity

Arulkarthick V. J, Dean IQAC and Professor,
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Abstract

Multimodal image fusion combines complementary information from diverse imaging sources, such as optical, infrared, and medical scans, to produce enhanced outputs for tasks like object detection and diagnosis. Traditional methods, including pixel-level and feature-level strategies, often struggle with artifacts, loss of modal-specific details, and complex data scenarios. This paper proposes a deep learning framework leveraging Convolutional Neural Networks (CNNs) for feature extraction and Generative Adversarial Networks (GANs) for adversarial training, enabling preservation of unique modal information, artifact reduction, and superior visual fidelity. Evaluations on medical (e.g., CT-MRI fusion) and remote sensing datasets demonstrate marked improvements in sharpness, contextual coherence, and resolution, outperforming conventional techniques. The approach addresses key challenges like low-resolution inputs and misalignment, paving the way for applications in autonomous systems and surveillance. Future work will explore hybrid architectures for real-time processing.

Keywords: *Multimodal image fusion; Convolutional Neural Networks (CNNs); Generative Adversarial Networks (GANs); contextual coherence*

MSEC-RS-2026-HR-005

Feasibility Analysis of Dredged Marine Sediment for Use in the Construction Industry

Anbu Neema G[1], Arivazhangan R[2]
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[2]Head Research and Professor, Department of Civil Engineering, Meenakshi Sundararajan Engineering College, Chennai 600 024, Tamil Nadu, India.
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Abstract

The increasing generation of dredged marine sediment (DMS) from port maintenance and navigation activities has raised significant environmental and disposal challenges. Conventional disposal methods, such as offshore dumping or confined storage, are not only costly but also pose serious ecological concerns. This study evaluates the feasibility of utilizing dredged marine sediment as a sustainable alternative to natural river sand in construction applications. Sediment samples were collected from Killai Fishing Harbour, Cuddalore District, Tamil Nadu, India, and subjected to comprehensive laboratory testing including sieve analysis, pH, organic matter content, specific gravity, direct shear, chloride and sulphate content, SEM analysis, and bulking characteristics, all in accordance with relevant Indian Standard (IS) codes. Results indicate that the sediment is predominantly sandy (95.57% sand) with low organic content (0.284%), a mildly alkaline pH (9.072), and acceptable shear strength (Φ = 26.16°). Comparisons with natural river sand show that most properties are within permissible IS code limits, with the exception of chloride content (0.22%), which necessitates washing or stabilization before use in reinforced concrete. SEM analysis confirmed angular particle morphology and dense packing, favorable for cement-based applications. The study concludes that DMS has strong potential as a sustainable construction material for concrete, mortar, road subgrades, embankments, and land reclamation.

Keywords: *Subgrade, Dredged Marine Sand, Construction Industry, Stabilization, Sustainability.*

MSEC-RS-2026-AI&DS-001

Swar Connect: Breaking Barriers, Connecting India

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Abstract

Linguistic differences can be challenging in interactions involving migrant workers, tourists, and other people coming from different regions speaking various regional Indian languages. Existing mobile applications do not allow real-time multilingual communication because of the absence of speech-to-speech capability. The paper proposes a deep learning-based voice communication and assistive system, named Swar Connect that provides speech-to-speech interaction in Indian languages. Swar Connect is able to perform real-time multilingual communication with ten different Indian languages utilizing advanced models such as IndicTrans, AI4Bharat language datasets, and eSpeakNG as the speech generator. It was developed using the flutter framework specifically for Android devices. Swar Connect recognizes spoken input in one language, translates it into another language, and synthesizes natural speech in real time. The system operates in low latency and has limited offline capabilities. Implementation results show higher efficiency, reduction of language barriers, and improved user experience. The contribution of Swar Connect to Industry 5.0 consists of creating human-centered intelligent communication systems for travel assistance, migrant support, education, etc.

Keywords: *Swar Connect, Speech-to-Speech Translation, IndicTrans, Multilingual AI, Voice Assistant, Industry 5.0*

MSEC-RS-2026-AI&DS-002

Artificial Intelligence in Decision Making for Public Policy and Governance

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Abstract

The integration of artificial intelligence into public sector decision-making represents one of the most consequential technological transitions of the 21st century. This paper examines the opportunities, risks, and governance frameworks arising from the deployment of AI systems in legislative drafting, judicial support, resource allocation, and administrative policy. Drawing on comparative case studies across democratic and transitional governance systems, we analyze how algorithmic tools are reshaping the relationship between state institutions and citizens. We further propose a tiered accountability architecture — grounded in principles of transparency, human oversight, and civic participation — to guide responsible adoption at national and sub-national levels.Artificial intelligence holds genuine promise to improve the quality, speed, and equity of public services. Realizing that promise, however, requires a governance architecture commensurate with the stakes involved. The challenge for democratic states is not merely technical — it is fundamentally political: how to embed human values, rights protections, and accountability mechanisms into systems whose inner workings resist easy scrutiny. The tiered accountability model proposed here offers a starting point, but its effectiveness will depend on sustained political will, adequate resourcing, and a genuine commitment to citizen participation in shaping the algorithmic state.

Keywords: *Federated Learning, Privacy-Preserving Voice Assistant, Edge Computing*

MSEC-RS-2026-AI&DS-003

A Hybrid Behavioral Analysis Framework Using Explainable Ai (Xai) For Real-Time Ransomware Detection

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Abstract

Ransomware has become a major cybersecurity threat to individuals, organizations, and infrastructures worldwide. They have their limitations in detecting evolving and zero-day ransomware attacks. In this paper, a hybrid framework for behavioral analysis of ransomware is proposed. It incorporates various static features, including file structure and entropy, and various dynamic features, including runtime behaviors, API calls, and system activities. The proposed model is based on a combination of Random Forest and Long Short-Term Memory (LSTM) techniques to effectively detect evolving and zero-day ransomware attacks. To make the proposed system transparent and trustworthy, various Explainable Artificial Intelligence (XAI) techniques, including SHAP, have been incorporated. The proposed system is tested using various benchmark datasets, including CIC-MalMem2022 and VirusShare. Additionally, deploying lightweight edge-based detection modules can support real-time protection in IoT and cloud environments. The use of automated response mechanisms (such as process isolation, rollback, and network quarantine) can further minimize damage. Enhancing XAI components with interactive visualization dashboards will also help security analysts understand attack patterns, trust model decisions, and perform faster incident investigations. Finally, continuous model retraining using live behavioral data and integration with SIEM/SOAR platforms can improve scalability, adaptability, and operational efficiency in enterprise cybersecurity environments. Future enhancements of the proposed framework include integrating real-time threat intelligence, lightweight and low-latency detection models, and cloud-native deployment for scalability. The system can be improved by incorporating reinforcement and federated learning to adapt to evolving ransomware attacks while preserving data privacy

Keywords *- Ransomware Detection, Explainable AI (XAI), SHAP, Hybrid Analysis, LSTM, Cybersecurity, Static Analysis, Dynamic Analysis, Real-Time Detection*

MSEC-RS-2026-AI&DS-004

A Unified Bert-Based Framework for Multimodal Fake News Detection with Contextual and Spread Analysis

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Abstract

The advancement and exponential growth of digital media has changed the way information is produced, distributed, or consumed. Although it has enhanced easier access to both data and communication, it has also led to the easy proliferation of incorrect news created in different forms of text, sound, or images. To identify the false news becomes a more difficult task in such a volatile environment, where surrounding context as well as dissemination behavior is to be taken into account. The proposed methodology is a multimodal deep learning framework for fake news detection using dual modality which combines semantic characteristic analysis based on contextual language understanding, audio feature extraction, and network based fake news spreading analysis model is proposed. This system approaches the problem in two different inputs, text modality and audio modality. The audio modality is converted into text by use of speech to text systems so that all inputs are in the text domain. The textual modality is then modeled by using the state- of-the-art pre-trained BERT model that helps to understand the context of the entire document.In order to increase interpretability and trust, we introduce a GNN module that captures the diffusion of news on a graph structure in the framework. This module informs the network how news generally Disseminate over the network for real and fake news as well as how it should. Moreover, we provide external references for the real news. We achieve enhanced accuracy and context knowledge on the data collected from Kaggle. Our method provides scalable and trustable answer to the multi-modal hate news.

Keywords: *Fake News Detection, Multimodal Learning, BERT, Speech Recognition, Graph Neural Networks, Contextual Information, Deep Learning.*

MSEC-RS-2026-AI&DS-005

FLAVA: Federated Learning Audio Voice Assistant for Privacy- Preserving Speech Recognition

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Abstract

Voice assistants play a central role in modern human–computer interaction by enabling hands-free operation across smartphones, smart speakers, wearables, and embedded systems. Their ability to interpret and respond to spoken commands has reshaped expectations for seamless and intuitive digital experiences. However, most current voice assistant architectures rely on centralized processing pipelines where raw audio or extracted features are transmitted to remote cloud servers for training and inference. This centralized model creates serious challenges, including privacy risks, potential unauthorized data access, and reliance on reliable network connectivity. The accumulation of sensitive speech data in central servers also raises ethical and regulatory concerns related to confidentiality and evolving data-protection standards. To overcome these limitations, this paper introduces FLAVA (Federated Learning Audio Voice Assistant), a privacy-preserving framework that decentralizes the learning process. FLAVA uses Federated Learning, allowing distributed model training directly on user devices so that raw audio never leaves local storage. Instead of sharing voice recordings, FLAVA transmits only model updates that are securely aggregated to create a global model capable of improving continuously while maintaining user privacy.

Keywords: *Federated Learning, Privacy-Preserving Voice Assistant, Edge Computing*

MSEC-RS-2025-AI&DS-006

DrLink AI – Intelligent Symptom Analysis & Smart Healthcare Connectivity

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Abstract

DrLink AI is an AI-enabled health care application that forecasts diseases based on the symptoms of users through NLP and machine learning models. It gives probabilistic predictions about possible health conditions. It also suggests appropriate specialists and doctors in the surroundings with the help of GPS functionality. This application helps in communicating with patients and doctors in real-time. Medical data sets are used for pre-processing and training the models. NLP extracts the most important symptoms provided by the users. The AI models predict the disease and provide the risk factors. Doctor mapping and GPS services assist users to access nearby medical practitioners. This technology incorporates AI prediction, doctor access, and a social networking service in the field of health care.

Keywords: *Artificial Intelligence, Symptom Analysis, Disease Prediction, Doctor Recommendation, Healthcare Chatbot*

MSEC-RS-2025-AI&DS-007

Smart E-Mail Prioritizer and Gmail Assistant

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Abstract

A Smart E-mail Prioritizer and Gmail Assistant is an intelligent system designed to enhance email management by automatically organizing, filtering, and prioritizing messages based on user behavior and contextual relevance. With the increasing volume of daily emails, users often struggle to identify important messages, leading to reduced productivity and missed opportunities. This system leverages machine learning and natural language processing techniques to analyze email content, sender importance, user interaction history, and urgency indicators. It classifies emails into categories such as high priority, promotional, social, and spam, ensuring that critical communications are highlighted and easily accessible.

The assistant integrates seamlessly with Gmail, providing features such as smart notifications, automated replies, and personalized email summaries. It can suggest responses, schedule emails, and remind users of pending actions, thereby acting as a proactive digital assistant. Additionally, the system continuously learns from user feedback to improve its accuracy and adaptability over time. By reducing manual effort and cognitive load, the Smart E-mail Prioritizer and Gmail Assistant significantly improves efficiency, enabling users to focus on meaningful tasks while maintaining effective communication management in both personal and professional contexts.

Keywords: *Spam Filtering, Digital Assistant, Automated Email Sorting, Gmail Assistant*

MSEC-RS-2026-AI&DS-008

Ethical Perspectives in Modern Artificial Intelligence Applications

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Abstract

Artificial Intelligence (AI) is rapidly reshaping modern society by enabling machines to perform complex tasks that typically require human intelligence. Despite its advantages, the increasing use of AI introduces critical ethical challenges that must be carefully managed. This study examines key ethical aspects of AI, focusing on fairness, transparency, accountability, privacy, and security**.**A major ethical concern is algorithmic bias, where AI systems may produce unfair outcomes due to biased or incomplete data. Addressing this issue requires careful data selection, continuous evaluation, and inclusive design practices. Transparency is equally important, as many AI models lack clarity in their decision-making processes. Promoting explainable AI helps users understand and trust these systems.In addition, AI systems often depend on large-scale data collection, raising serious privacy concerns. Ensuring proper data protection and responsible usage is essential to safeguard individuals' rights. Moreover, ethical AI development should aim to benefit society while minimizing harm. This includes addressing potential risks such as job displacement, misuse of AI applications, and unintended consequences. Collaborative efforts between policymakers, researchers, and industry professionals are necessary to create effective ethical guidelines and regulations. In conclusion, integrating ethical principles into AI development is essential for building reliable and trustworthy systems. By prioritizing responsible practices, AI can be used to enhance human well-being while reducing potential risks.

Keywords: *Artificial Intelligence, Ethics, Bias, Transparency, Accountability, Explainable AI, Data Security, Responsible AI*

MSEC-RS-2026-AI&DS-009

A Robust Machine Learning Framework for Early Disease Prediction and Risk Assessment

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Abstract

Early detection of diseases and accurate risk assessment are critical for improving patient outcomes and reducing healthcare costs. This study proposes a robust machine learning framework for early disease prediction and risk assessment using clinical and demographic data. The proposed system integrates data preprocessing, feature selection, and multiple machine learning algorithms to identify patterns associated with disease onset. Models such as Logistic Regression, Support Vector Machine, Random Forest, and Gradient Boosting are evaluated to determine the most effective predictive approach. To enhance model performance and reliability, techniques such as normalization, handling missing values, and hyperparameter tuning are employed. The framework also incorporates risk stratification to categorize patients into different risk levels, enabling timely medical intervention. Experimental results demonstrate that the proposed approach achieves high accuracy, precision, recall, and F1-score compared to traditional methods. Furthermore, the model emphasizes interpretability, allowing healthcare professionals to understand key contributing factors influencing predictions. The proposed system can serve as a decision-support tool for clinicians, facilitating early diagnosis and improving preventive healthcare strategies. This work highlights the potential of machine learning in transforming healthcare through data- driven insights and proactive risk management.

Keywords: *Machine Learning, Early Disease Prediction, Risk Assessment, Healthcare Analytics, Predictive Modeling, Feature Selection*

MSEC-RS-2026-AI&DS-010

Leveraging Deep Learning for Predictive Modeling in Clinical Research and Pharmaceutical Discovery

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Abstract

The traditional drug discovery process is notoriously slow, expensive, and inefficient, with nearly 90% of candidates failing after entering phase-I clinical trials. This paper argues that deep learning (DL) represents a fundamental paradigm shift, offering a transformative solution to these persistent challenges. We demonstrate how DL frameworks are revolutionizing core tasks, from accelerating disease diagnosis and target identification to optimizing virtual screening and predicting drug toxicity with greater accuracy. By leveraging models that learn directly from complex, high-dimensional data—including genomic sequences, molecular structures, and electronic health records—DL uncovers critical patterns that elude conventional methods. The primary impact is a significant de-risking of the therapeutic pipeline, enhancing the probability of success while drastically reducing development timelines. We conclude that the integration of these advanced computational tools is essential for fostering innovation and realizing the future of data-driven, personalized healthcare.

Keywords: *Deep Learning; Drug Discovery; Virtual Screening; Drug Toxicity Prediction; Personalized Healthcare; Therapeutic Pipeline Optimization.*

MSEC-RS-2026-AI&DS-011

An Intelligent NLP-Based Framework for Automated Resume Screening and Candidate Filtering

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Abstract

The exponential increase in job applications has made manual resume screening a time-consuming and error-prone process in modern recruitment systems. This paper presents an intelligent Natural Language Processing (NLP)-based framework for automated resume screening and candidate filtering. The proposed system integrates advanced text pre-processing, feature extraction using Term Frequency–Inverse Document Frequency (TF-IDF), and semantic similarity measurement through cosine similarity to evaluate candidate profiles against job descriptions. Unlike traditional keyword-matching approaches, the framework captures contextual relevance and domain-specific semantics, enabling more accurate candidate ranking. The system further incorporates machine learning-based classification to enhance decision-making efficiency and reduce bias in recruitment. Experimental evaluation on benchmark and real-world datasets demonstrates significant improvements in precision, recall, and overall screening accuracy compared to conventional methods. The architecture is designed to be scalable and deployable in real-time recruitment environments, supporting large volumes of applications. Additionally, the framework provides decision-support insights to recruiters, facilitating faster and more reliable hiring processes. Future work will explore transformer-based language models, explainable AI techniques, and adaptive learning mechanisms to further improve system robustness and transparency.

Keywords: *NLP; Resume Screening; TF-IDF; Cosine Similarity; Machine Learning, Recruitment Automation; Semantic Analysis*

MSEC-RS-2026-AI&DS-012

A Smart Task Management and Monthly Planning Application with AI Assistance and Realtime Collaboration

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Abstract

In today's fast-paced digital environment, effective task management and time planning are essential for improving productivity and maintaining work-life balance. This project presents the design and development of a Smart Task Management and Monthly Planning Application integrated with Artificial Intelligence (AI) assistance and real-time collaboration features. The application aims to help users efficiently organize, prioritize, and track their daily tasks and monthly goals through an intuitive and user-friendly interface. The system leverages AI algorithms to provide intelligent task recommendations, deadline predictions, priority adjustments, and personalized productivity insights based on user behaviour and historical data. Additionally, it supports real-time collaboration, enabling multiple users to share task lists, assign responsibilities, and communicate seamlessly within teams. This makes the application suitable for both individual users and collaborative work environments. Key features include dynamic task scheduling, automated reminders, progress tracking, calendar integration, and secure cloud-based data synchronization. The monthly planning module allows users to visualize their long-term objectives and break them down into manageable tasks. The integration of AI enhances decision-making and reduces manual effort, while real-time updates ensure that all collaborators stay aligned. Overall, this application provides a comprehensive solution for modern task management challenges by combining intelligent automation with collaborative efficiency, ultimately improving productivity and goal achievement.

Keywords: *Smart Task Management, Monthly Planning, Artificial Intelligence (AI), Real-Time Collaboration, Productivity Enhancement,*

MSEC-RS-2026-AI&DS-013

An Intelligent Community Carbon Exchange Platform Using Artificial Intelligence for Incentivizing Sustainable Actions

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Abstract

The impact of climate change is getting worse and worse. This means we really need to reduce the things we do to the environment. We need to encourage people to be kind to the earth. This is a proposal for a platform called Community Carbon Exchange. It is a website where people can get rewards for doing things for the environment. People can use this platform to log the things they do every day. For example, they can log when they ride a bike, reduce waste or use energy. The platform uses computer programs to check if people are really doing these things. If they are, they get credits. These credits are like money that they can use to buy things or donate to people who help the environment. They can even trade these credits with their friends. The platform also has games and challenges to make it fun for people to help the environment. It is like a competition where people can see who is doing the job of helping the earth. The platform also uses computer programs to give people advice on how they can do better. Community Carbon Exchange is an idea because it helps people work together to help the environment. It makes helping the earth a fun and exciting thing to do. The platform can be used by anyone. Can be expanded to many people. It can really help us achieve our goals of making the earth a happier place. The Community Carbon Exchange platform is very good at helping people be kind to the earth. It gives people rewards for doing things and helps them work together to make a difference. The Community Carbon Exchange is a platform that can make a big difference.

Keywords: *Artificial Intelligence (AI), Community Carbon Exchange, Sustainability, Carbon Credits, Machine Learning, Predictive Analytics, Smart Communities, Green Technology.*

MSEC-RS-2026-AI&DS-014

Combating Poisoning Attacks in Blockchain-Enabled Federated Learning in Healthcare

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Abstract

Federated Learning is a decentralized machine learning approach that enables multiple entities to collaboratively train a model without sharing raw data, ensuring data privacy and security. The related literature has shown that the scope of Federated Learning and Blockchain in various fields are studied by many researchers. Blockchain is a distributed ledger with growing lists of records (blocks) that are securely linked together via cryptographic hashes. Each block contains a cryptographic hash of the previous block, a timestamp, and transaction data generally represented as a Merkle tree, where data nodes are represented by leaves. Since each block contains information about the previous block, they effectively form a chain (viz. linked list data structure), with each additional block linking to the ones before it. Federated Learning enables collaborative model training while preserving data privacy. This research article evaluates the impact of malicious clients on FL and explores blockchain integration as a defence mechanism to enhance security and transparency. Experimental results show that in the presence of malicious clients, the global model achieves an accuracy of 84%, whereas excluding the malicious ones by our reward and slash mechanism improves accuracy to 93%. A reward-slash mechanism is implemented to penalize low-performing or malicious participants, to ensure reliable client contributions. Validator selection based on token and accuracy scores further strengthens model integrity, while blockchain-based auditability ensures tamper-proof validation of updates. The findings highlight how blockchain integration enhances FL's robustness against adversarial threats, making it a viable solution for privacy-sensitive applications such as healthcare.

Keywords: *Federated Learning, Blockchain, FedAVG, Smart Contracts, IPFS, Reward and Slash Mechanism.*

MSEC-RS-2026-CE-001

Buoyant Behaviour of Sustainable Hydro Housing

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Abstract

This study investigates the floating performance of a sustainable hydro housing prototype intended for temporary emergency shelter applications. The developed system incorporates sealed plastic containers as buoyant units combined with a lightweight structural platform, enabling economical and environmentally responsible construction. Experimental evaluation was carried out under controlled conditions to assess performance indicators including immersion depth, and buoyant force. These results demonstrate that the proposed hydro housing system is a practical and cost-effective solution for short-term flood resilience. The study demonstrates that the proposed hydro housing sys-tem is a feasible, low-cost, and sustainable solution for temporary flood-resilient applications

Keywords: *Buoyant, Sustainable, hydro, floating, resilient, housing*

MSEC-RS-2026-CE-002

Seismic Fragility Analysis of Vertically Irregular Steel Framed Buildings

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Abstract

Disaster mitigation of the country requires the data of fragility curves for various typical buildings. As the number of steel buildings in the cities is increasing, development of seismic fragility curves of steel buildings are also the need of the hour. This study discusses the fragility curves and damage index functions of a 7-storey regular and irregular steel-framed buildings subjected to different earthquake ground motions. Each building was designed based on the codes, IS 800 (2007) and IS 1893 part 1 (2016).

Time History Analysis of various vertically irregular framed buildings was performed and displacements of each floor were obtained. Inter-storey drift ratio and spectral acceleration at fundamental period are taken as damage parameter and intensity measure for deriving the fragility curves. The probabilistic seismic demand models for typical stiffness and mass vertical irregular buildings were developed. Steel framed buildings with ground storey open were found to be the vertically irregular building which is the most vulnerable out of all the selected vertical buildings.

Keywords: *Seismic Fragility Curves, Steel Frame Structures, Vertical Irregularity, Inter-storey Drift Ratio, Time History Analysis.*

MSEC-RS-2026-CE-003

Volatile Organic Compounds Emission Analysis from Traffic Signals in Chennai

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Abstract

Volatile organic compounds (VOCs) are ubiquitous domestic pollutants. Their role in asthma/allergy development and exacerbations is uncertain. This systematic review investigated whether domestic VOC exposure increases the risk of developing and/or exacerbating asthma and allergic disorders. We systematically searched 11 databases and three trial repositories, and contacted an international panel of experts to identify published and unpublished experimental and epidemiological studies. 8455 potentially relevant studies were identified; 852 papers were removed after de-duplication, leaving 7603 unique papers that were screened. Of these, 278 were reviewed in detail and 53 satisfied the inclusion criteria. Critical appraisal of the included studies indicated an overall lack of high-quality evidence and substantial risk of bias in this body of knowledge. Aromatics (i.e. benzenes, toluene's and xylenes) and formaldehyde were the main VOC classes studied, both in relation to the development and exacerbations of asthma and allergy. Approximately equal numbers of studies reported that exposure increased risks and that exposure was not associated with any detrimental effects. The available evidence implicating domestic VOC exposure in the risk of developing and/or exacerbating asthma and allergy is of poor quality and inconsistent. Prospective, preferably experimental studies, investigating the impact of reducing/eliminating exposure to VOC, are now needed in order to generate a more definitive evidence base to inform policy and clinical deliberations in relation to the management of the now substantial sections of the population who are either at risk of developing asthma/allergy or living with established disease.

Keywords: *Volatile organic compounds; Aromatic compounds; Indoor air pollutants, Sensitisation.*

MSEC-RS-2026-CE-004

Simulation-Driven Evaluation of an Adaptive Edge-Cloud Prescriptive IoT Framework for Mode-Share-Aware Safety Enhancement at Non-Signalized Intersections

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Abstract

Non-signalized intersections in mixed traffic environments exhibit high crash susceptibility due to the absence of control mechanisms and heterogeneous vehicle interactions. This study proposes a simulation-driven adaptive edge–cloud prescriptive framework using modular IoT nodes to enhance intersection safety through real-time risk estimation and mode-share differentiation. Unlike conventional approaches, the framework integrates edge-level sensing and classification of road users (two-wheelers, cars, pedestrians) with cloud-based predictive analytics to dynamically compute a risk index based on traffic headway, conflict probability, and temporal flow variations. A digital twin of the intersection is developed using an online simulation platform, where IoT nodes are emulated to capture distance, speed, and density parameters. The system employs rule-based and data-driven logic to prescribe adaptive interventions such as virtual warnings and priority allocation. Experimental scenarios with varying traffic compositions demonstrate that the proposed framework significantly reduces conflict probability and improves safety performance compared to static conditions. The results highlight that mode-share-aware safety modeling is critical in heterogeneous traffic contexts, with two-wheelers contributing disproportionately to risk under high-density conditions. The study establishes that low-cost simulation tools can effectively validate IoT-based safety frameworks, offering a scalable pathway for real-world deployment in developing urban environments.

Keywords: *Internet of Things (IoT); Edge–Cloud Computing; Non-Signalized Intersections Traffic Safety Analytics; Mode-Share Differentiation; Risk Index Modelling; Digital Twin Simulation*

MSEC-RS-2026-CE-005

Adaptive Web App for Runoff Estimation Using SCS-CN and Stranger's Method

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Abstract

Surface runoff estimation plays a crucial role in hydrology and water resource management, as it aids in understanding the impact of rainfall on watershed behaviour. This developed project, titled RunMeter – Runoff Estimation App, provides automated runoff estimation and hydrograph visualization using Python, Streamlit, and GitHub integration for estimating runoff efficiently using two empirical approaches: the SCS-CN (Soil Conservation Service Curve Number) Method and the Strange's Method. The web app allows users to upload rainfall data, input parameters such as Curve Number (CN) or Runoff Coefficient (C), and automatically compute runoff volume. It further generates hydrographs and tabulated discharge data for analysis and visualization. The system aims to simplify hydrological computations and make runoff estimation accessible to students, researchers, and professionals. Future enhancements include integrating real-time location-based rainfall data, and adding a live satellite map to alert users about potential high runoff zones. The adaptive framework of the web app ensures an efficient, accurate, and user-friendly platform for surface runoff prediction and water resource assessment.

Keywords: *RunMeter, Curve Number, Runoff Coefficient, SCS-CN, rainfall data, watershed.*

MSEC-RS-2026-CE-006

A Study on Thermal Performance and Comfort of a Multi- Storied Building

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Abstract

The population growth is enormous. But the available residential buildings are insufficient. The requirement has broken the construction field to build multistoried buildings. Agricultural lands have been utilized to construct high-rise buildings. A High-rise building utilizes a lesser land area than the area used by the individual homes. In a multi-storey building, the roof of one flat becomes the floor of the other, reducing the construction cost. In this type of construction, symmetry and elegance exists. The doors, windows, and, ventilators are provided in the same size. The mono color is used for the outer walls and another mono color is applied to the inner walls of the flats except the design items. The problem arising for the apartments is the utilization of more energy and water. Plenty of steel is incorporated for the construction from the basement to the top floor. Steel exhibits immediate thermal responses which influence the inner space of the building. Consequently, it disturbs the pleasant thermal condition. The rainy and winter seasons do not bring much change. But the summer season transfers sun radiations into the interior. Based on the material used for the construction, heat is transferred into the inside through conducting materials, and also through convection and radiation. The study aims to carry out the heat changes found in the walls, roof, and floor of a test building during diurnal hours, and to find out whether it satisfies the comfort level or not.

Keywords: *Roof temperature, Wall temperature, Indoor temperature, thermal performance, Comfort, Heat transfer, Multi-storey buildings.*

MSEC-RS-2026-CE-07

Lead-Free Gamma Radiation Shielding Geopolymers Incorporating Nano-Bismuth Oxide: Mechanical Performance, Microstructural Densification, and Attenuation

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Abstract

This study presents the development of a lead-free geopolymer-based radiation shielding composite incorporating nano-bismuth oxide (Bi_2O_3) as a high atomic number filler. Class F fly ash geopolymers were synthesized using alkaline activation, with varying percentages of nano-Bi_2O_3 to enhance gamma-ray attenuation while maintaining mechanical performance. Microstructural analysis confirmed improved matrix densification due to effective particle packing and pore refinement. Mechanical tests showed that optimal nano-Bi_2O_3 content maintained or slightly enhanced compressive strength. Gamma attenuation properties, including linear and mass attenuation coefficients, half-value layer, and radiation protection efficiency, demonstrated significant improvement with increasing nano-Bi_2O_3 content, especially at low to medium photon energies. Compared to conventional lead- and barite-based materials, the developed composite offers competitive shielding performance without toxicity concerns, making it a sustainable and efficient alternative for applications in nuclear facilities, medical imaging, and radioactive waste containment.

Keywords: *Geopolymer concrete; Bismuth oxide nanoparticles; Gamma radiation shielding; Lead-free shielding materials; Attenuation coefficient; Sustainable construction materials*

MSEC-RS-2026-CE-008

Analytical and Numerical Study of RC Deck Slab Under IRC Class AA Loading

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Abstract

This project focuses on the analysis and design of a reinforced concrete deck slab for a flyover bridge. The deck slab is one of the most critical structural components, as it directly carries vehicular loads and transfers them safely to the supporting elements such as girders, piers, and foundations. Its proper design is essential to ensure structural stability, durability, and user safety under varying traffic and environmental conditions. The proposed flyover has a total length of 3 km, consisting of 150 spans with each span measuring 20 m. Due to its large scale and continuous loading conditions, careful consideration is given to load distribution, bending moments, shear forces, and deflection characteristics of the deck slab. The manual design of the deck slab is carried out in accordance with relevant Indian Standard codes, including IS: 21-2000 and IS:452-2000, ensuring compliance with prescribed safety and design guidelines. Various loads such as dead load, live load and impact load are considered in the design process. In addition to manual calculations, structural analysis is performed using STAAD Pro Connect Edition software. The software is utilized to model the deck slab and simulate real-time loading conditions, enabling accurate evaluation of structural behavior. The results obtained from the software are compared with manual calculations to verify accuracy and reliability. This project emphasizes the integration of conventional design methods with modern software tools to achieve an efficient, safe, and economical design of flyover deck slabs.

Keywords: *Flyover Bridge, Reinforced Concrete Deck Slab, Structural Analysis, STAAD Pro Connect Edition*

MSEC-RS-2026-CE-009

Performance Enhancement of Coconut Waste Materials in Concrete: A Systematic Review with Emphasis on Nano-Treatment and Durability

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Abstract

Coconut waste materials derived from *Cocos nucifera* offer significant potential for sustainable concrete; however, their large-scale adoption is limited by durability issues and weak fibre–matrix interactions. This systematic review synthesises 142 peer-reviewed studies (1960–2026) to evaluate the performance of coir fibre, coconut shell aggregate, shell ash, coir pith, and hybrid systems, with a focused emphasis on nano-engineering strategies. A key contribution of this study is the development of a multiscale framework linking nano-modification to macro-level concrete performance**.** Coir fibre enhances toughness and impact resistance but is highly susceptible to alkaline degradation. Nano-treatment approaches, including nano-silica, nano-clay, and nano-lime, significantly improve performance through C–S–H gel densification, pore refinement, and strengthening of the interfacial transition zone (ITZ)**.** These mechanisms reduce permeability, improve fibre–matrix bonding, and enhance durability. Coconut shell aggregate is suitable for lightweight concrete, while coconut shell ash exhibits strong pozzolanic activity. Nano-modification further enhances hydration kinetics and promotes secondary C–S–H formation, leading to improved mechanical and durability properties. Technology readiness assessment indicates that most nano-modified systems remain at laboratory to pilot scale (TRL 3–5), highlighting a gap in field validation. The review identifies critical research needs, including optimisation of nano-treatments, long-term performance studies, and standardisation frameworks for practical implementation.

Keywords: *Nano-engineered concrete; Coconut waste materials; Nano-silica; Interfacial transition zone (ITZ); C–S–H densification; Durability enhancement; Sustainable construction; Hybrid composites.*

MSEC-RS-2026-CE-010

Multi-Criteria Optimization of High-Strength Ambient-Cured GGBS–Metakaolin Geopolymer Binders Using TOPSIS

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Abstract

The escalation of greenhouse gas emissions has driven the development of "low carbon binders," with geopolymer concrete emerging as a prominent eco-friendly alternative to traditional cement. This study investigates the development of high-strength geopolymer binders optimized for ambient curing conditions. Despite their environmental benefits, the practical application of geopolymers is often hindered by the complexity of selecting appropriate mix proportions from numerous influential parameters. To address this, a Multi-Criteria Decision-Making approach using the TOPSIS method was employed to evaluate and optimize binder performance. Experimental mixtures were prepared using Ground Granulated Blast-furnace Slag and metakaolin as precursors. Thirty-two mortar combinations were synthesized by varying the metakaolin percentage, alkaline activator ratio (sodium silicate to sodium hydroxide), and sodium hydroxide concentration. The performance of these mixtures was assessed based on nine response factors: flow characteristics, initial and final setting times, compressive strength (at 7 and 28 days), initial drying shrinkage, energy consumption, CO_2 emissions, and cost-effectiveness. Additionally, microstructural analysis was conducted to determine the impact of MK on the GGBS-based geopolymer matrix. The application of multi-response optimization techniques in this study provides a systematic framework for achieving desired material attributes, potentially enhancing the feasibility and utilization of geopolymer concrete in the construction industry.

Keywords: *Geopolymer concrete, Low carbon binders, GGBS, Metakaolin, TOPSIS, Multi-Criteria Decision-Making, Ambient curing, Optimization.*

MSEC-RS-2026-CSE-001

Detecting Deepfake Harassment Using Digital Forensics

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Abstract:

The rapid democratization of generative artificial intelligence (AI) has precipitated a severe escalation in synthetic media weaponization, transforming deepfake-enabled harassment into a critical, high-impact cybersecurity threat. Existing mitigation strategies predominantly rely on probabilistic machine learning (ML) classifiers, such as Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). However, these approaches are computationally expensive, highly vulnerable to adversarial evasion techniques, and critically lack the deterministic explainability required for legal admissibility in judicial proceedings. This paper proposes a fundamental paradigm shift from heuristic authenticity prediction to deterministic integrity verification. We introduce Evidence Media, an integrity-first, CPU-optimized digital forensics framework designed to analyze media provenance without relying on AI classification. The implemented prototype executes a multi-layered forensic pipeline utilizing SHA-256 cryptographic hashing for an immutable chain of custody, exhaustive metadata extraction via ExifTool, and structural container analysis using FFmpeg within a secure, privacy-preserving web architecture. Experimental evaluation across pristine, structurally tampered, and AI-generated synthetic datasets demonstrates a 100% reproducibility rate in anomaly detection and zero hash collisions. The core contribution of this work is a scalable, legally defensible verification tool that empowers victims of digital gender-based violence and cyber harassment with automated, human-readable forensic reports, completely bypassing the adversarial vulnerabilities of traditional AI-based detection.

Keywords: *Deepfake Harassment, Cybersecurity, Digital Forensics, Media Integrity, Provenance Tracking, Cryptographic Verification*

MSEC-RS-2026-CSE-002

Predictive Cyber Threat Analysis Using Machine Learning

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Abstract:

In today's digital era, malware continues to pose a significant threat to computer systems and networks, often exploiting misconfigured or weak system environments. Traditional malware detection techniques largely rely on signature-based methods, which struggle to detect new or evolving malware variants. This paper proposes a machine learning–based approach for predicting the likelihood of malware infection based on a system's configuration and specifications. A labeled dataset comprising various system attributes—such as RAM size, CPU cores, operating system version and security settings—was used to train a classification model. The Random Forest algorithm was employed due to its robustness and high accuracy in handling structured data. The trained model is capable of classifying whether a system is likely to be vulnerable to malware or not. To further validate the model's effectiveness, this framework includes a simulation environment using a virtual machine, where specific system configurations can be emulated to test real-world malware behavior. Additionally, a front-end interface has been developed to allow users to input system specifications and receive instant predictions. This approach not only enhances proactive security assessment but also helps identify high-risk configurations before malware exploitation occurs. The integration of predictive modeling with Virtual Machine-based simulation serves as a practical step toward more intelligent and adaptive cybersecurity systems.

Keywords: *Malware; Threat; Virtual Machine; Random Forest Algorithm, Machine Learning*

MSEC-RS-2026-CSE-003

Early Hate Onset Detection in Online Conversations

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Abstract:

Online conversational platforms increasingly require intelligent systems capable not only of detecting hate speech but also of identifying the point at which hateful intent first emerges during an interaction. This study proposes a novel framework for hate onset detection that determines the earliest conversational turn where hateful content is introduced. Conversations are represented as ordered sequences of utterances and encoded using language-agnostic embeddings to capture semantic relationships independent of linguistic variations. To improve interpretability and model robustness, the framework incorporates Explainable Artificial Intelligence (XAI) techniques that analyze prediction behavior and identify influential hate-related features, which are then used for feature refinement. Experimental evaluation on the Gab conversational dataset demonstrates that the proposed approach achieves an accuracy of 94%, a Macro F1-score of 0.75, and a Weighted F1-score of 0.94. Compared with the baseline model without feature refinement, the proposed method yields a 3% improvement in accuracy, a 0.24 increase in Macro F1-score, and a 3% gain in Weighted F1-score.

Keywords: *Hate Onset Detection; Language-Agnostic Embeddings; Explainable Artificial Intelligence (XAI); Conversational Hate Speech Analysis; Early Inten.*

MSEC-RS-2026-CSE-004

Self-Supervised Underwater Image Enhancement via Attention-Guided CNN with PPDAT

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Abstract:

Underwater object detection and classification are severely challenged by the poor visual quality of underwater images caused by wavelength-dependent light absorption, scattering, and non-uniform illumination. These effects lead to color distortion, low contrast, and reduced visibility, significantly degrading the performance of conventional computer vision models. To address this problem, this paper presents a self-supervised underwater image enhancement framework that improves visual quality without relying on paired ground-truth images. The proposed approach employs an attention-guided convolutional neural network, integrating both channel-wise and spatial attention mechanisms to effectively focus on degraded regions and salient features. In addition, physical priors of underwater image formation, including transmission characteristics and color attenuation properties, are incorporated to guide the enhancement process in a physically consistent manner. A degradation-aware self- supervised training strategy is further introduced by enforcing consistency between the enhanced image and its synthetically re-degraded version, enabling robust learning from unpaired underwater datasets. Experimental evaluation using standard perceptual and fidelity metrics such as UIQM, UCIQE, PSNR, and SSIM demonstrates significant improvements in color restoration, contrast enhancement, and structural preservation. The proposed method serves as an effective preprocessing step for downstream underwater vision tasks, including object detection and classification.

Keywords: *Underwater image enhancement, convolutional neural network, degradation-aware, attention mechanisms, degradation-aware*

MSEC-RS-2026-CSE-005

Self-Healing Federated IOT Network with Trust & Adaptive Defense

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Abstract:

The rapid expansion of Internet of Things (IoT) devices has intensified concerns regarding security, privacy, and data integrity. While Federated Learning (FL) enables decentralized model training without sharing raw data, it remains vulnerable to malicious devices that can compromise model reliability. This paper proposes a Self-Healing Federated IoT Network with Trust and Adaptive Defense, aimed at enhancing robustness in distributed IoT environments. The proposed framework introduces a dynamic trust management mechanism that assigns trust scores to devices based on their behavioral patterns. Devices exhibiting anomalous or malicious activity are penalized and isolated once their trust score falls below a defined threshold. To strengthen detection, lightweight anomaly detection techniques combining statistical methods and machine learning are employed to identify irregular data contributions in real time. A key feature of the system is its self-healing capability, which reconstructs the global model using only trusted devices following an attack. Additionally, an attack memory module stores historical attack zcloud architecture ensures scalability and low-latency processing. The proposed system enhances the security, adaptability, and reliability of IoT networks in critical applications.

Keywords: *Federated Learning, IoT Security, Trust Management, Self-Healing Systems, Anomaly Detection, Edge Computing*.

MSEC-RS-2026-CSE-006

Secure Messaging Website with Time Bound Encrypted File Sharing

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Abstract:

In modern digital communication, protecting sensitive information during transmission has become increasingly critical due to the growing risks of data breaches, cyberattacks, and unauthorized access. This project proposes a secure web-based messaging system designed to ensure confidentiality, integrity, and controlled accessibility of shared information. The platform enables users to send messages and files that are encrypted before transmission, ensuring that only intended and authenticated recipients can decrypt and access the data. Authentication mechanisms such as secure tokens, passwords, or time-based access keys are used to verify user identity and prevent unauthorized entry. A key feature of the proposed system is its time-restricted file-sharing capability. Unlike traditional file-sharing platforms that store data indefinitely, this system allows the sender to define a specific access duration for each shared file. Once the defined time limit expires, the file is automatically deleted from the system, thereby reducing the risk of long-term data exposure, misuse, or unauthorized redistribution. In addition to encryption and access control, the platform incorporates advanced security measures to enhance user safety. All messages, attachments, and shared links are scanned using integrated threat detection mechanisms to identify potential malware, phishing attempts, or malicious content. If any suspicious activity is detected, access to the content is immediately restricted, and appropriate alerts are generated to protect users from potential harm. This approach not only strengthens data privacy but also builds user trust by ensuring that sensitive information is shared in a controlled, secure and time-bound environment.

Keywords: *Secure Messaging System, End-to-End Encryption, Time-Based Access Control, Secure File Sharing, Cybersecurity, Threat Detection, Data Integrity, Confidential Communication.*

MSEC-RS-2026-CSE-007

MEDICHAIN: A Causal-Aware Multi-Modal AI Framework for Intelligent Timeline Reconstruction in Digital Forensic Investigation

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Abstract:

Digital forensic investigation is essential for understanding the sequence of events from various types of large-scale system data. However, current methods for creating timelines mainly depend on timestamps and rules to connect events, which fail to capture the relationships and cause-and-effect links between events. This can lead to incomplete or incorrect conclusions, especially in complex situations with data from multiple sources like system logs, user actions, and interactions with IoT devices. To solve these issues, this paper introduces a Causal-Aware Multi-Modal Artificial Intelligence Framework for intelligent timeline reconstruction in digital forensics. A temporal sequence learning model, based on LSTM networks, is used to understand how events unfold over time, The framework also includes an event importance scoring system to highlight key forensic actions and builds a directed event graph that shows both the time order and cause-effect connections of events. The framework was tested using both simulated and real-world log data, and the results showed that this approach greatly improves the accuracy and coherence of timeline reconstruction compared to traditional methods. It also enhances the ability to find meaningful event sequences and cause-effect relationships, reducing the need for manual investigation. This framework advances intelligent digital forensics by incorporating causal reasoning, multi-modal data integration, and explainable AI, providing a scalable and effective solution for future forensic analysis systems.

Keywords: *Digital Forensics, Timeline Reconstruction, Causal Inference, Multi-Modal Data Fusion, Artificial Intelligence, LSTM, Event Correlation, Explainable AI, Forensic Analytics, Event Graph Modelling*

MSEC-RS-2026-CSE-008

Crime pattern analysis and Prediction system using Machine Learning

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Abstract:

Crime has become a major concern in modern society due to rapid urbanization, population growth, and socio-economic changes. Traditional crime analysis methods rely heavily on manual investigation and historical record examination, which are time-consuming, inefficient, and prone to errors. With the increasing availability of large-scale crime data, there is a growing need for intelligent and automated systems that can analyze crime patterns and predict future occurrences effectively. The Crime Pattern Analysis and Prediction System is designed to analyze historical crime datasets, identify hidden patterns, detect crime hotspots, and predict future crime trends using machine learning techniques. The system collects crime data from reliable sources and performs comprehensive data preprocessing, including data cleaning, normalization, and feature extraction. Various machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine are implemented to build predictive models. Among these, the Random Forest algorithm provides higher accuracy and robustness in handling complex and nonlinear datasets. The proposed system also incorporates interactive data visualization dashboards that present crime distribution based on time, location, and crime category. These visual insights enable law enforcement agencies to make informed decisions, optimize resource allocation, and plan preventive strategies efficiently. Experimental results demonstrate that the proposed system achieves high prediction accuracy and significantly improves crime analysis efficiency.

Keywords: *Random Forest, Data Visualization, Crime Hotspots, Predictive Modelling, Smart Policing.*

MSEC-RS-2026-CSE-009

AI-Driven Intelligent Workforce Management System Using Supervised Learning Algorithms

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Abstract:

The AI-Integrated Employee Management System is designed to enhance workforce management by replacing traditional manual methods with an intelligent and automated solution. Conventional employee management systems often involve fragmented processes, manual record-keeping, and time-consuming administrative tasks, which lead to inefficiencies and errors. This system provides a centralized, web-based platform that integrates multiple functionalities such as attendance management, task tracking, promotion evaluation, food poll management, and employee wellness monitoring, enabling organizations to manage their workforce more effectively. A key feature of the system is employee churn prediction, which uses supervised learning algorithms to analyze historical employee data and identify employees at risk of leaving. This predictive capability helps organizations take proactive measures to improve employee retention and overall productivity while also supporting data-driven decision-making. The system is developed using a modern full-stack architecture to ensure scalability, security, and performance. The frontend is built using React.js and Tailwind CSS for a responsive and user-friendly interface, while the backend uses Node.js and Express.js to provide secure and efficient server-side processing. Overall, the system serves as a comprehensive and future-ready solution for intelligent workforce management, with potential for further enhancements such as advanced analytics and AI-driven decision support.

keywords: *Employee Management; Supervised learning algorithms; Workforce Automation; Employee Churn Prediction; Predictive Analytics Human Resource; Attendance; Full-Stack; Web Application;*

MSEC-RS-2026-CSE-010

Adaptive Explainable AI for Real-Time Decision Systems

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Abstract:

Artificial Intelligence (AI) systems are increasingly deployed in real-time decision-making environments such as healthcare monitoring, autonomous vehicles, and financial trading. However, the lack of transparency and adaptability in many AI models limits their reliability and user trust. This work proposes an Adaptive Explainable AI (AXAI) framework that integrates dynamic learning capabilities with interpretable model outputs. The framework combines reinforcement learning for continuous adaptation with explainable techniques such as attention mapping and rule extraction to provide human-understandable insights into model decisions. Unlike traditional static models, AXAI evolves based on changing input patterns while maintaining a transparent reasoning process. Experimental evaluation demonstrates improved accuracy, faster response time, and enhanced user trust compared to conventional black-box models. This approach bridges the gap between performance and interpretability, making it suitable for high-stakes applications where accountability and clarity are critical.

Keywords: *Adaptive AI, Explainable AI (XAI), Real-Time Decision Systems, Reinforcement Learning, Model Interpretability, Dynamic Learning, Human-AI Interaction, Transparency, Trustworthy AI*

MSEC-RS-2026-CSE-011

Optimized Implementation of Face Recognition Algorithms for Real-Time Surveillance and Identity Verification

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Abstract:

The Face recognition is to category of biometric security. The project explores the implementation of a face recognition system using Python and computer vision techniques. The system aims to accurately identify individuals from images or video streams by comparing their facial features against a database of known faces. This technology finds applications in various domains, including security systems, access control, and human-computer interaction. The project focuses on building a robust and efficient face recognition system, addressing challenges such as variations in lighting, pose, and facial expressions. Face recognition system is very essential and important for providing security, mug shot matching, law enforcement applications, user verification, user access control, etc and is mostly used for recognition for various applications.

Keywords: *Face Recognition, Real-Time Surveillance, Identity Verification, Deep Learning, Optimization, Edge Computing, Computer Vision*

MSEC-RS-2026-CSE-012

NAMMA TRANSIT: One Tamil Nadu. One App. For Everyone

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Abstract:

Regions such as Tamil Nadu undergoes rapid urbanization and population growth, this has significantly increased the complexity of public transportation systems. Fragmented services, overcrowding, inefficient route planning and limited accessibility for diverse user groups are the common challenges that arise. Existing transportation applications are all separately divided based on transportation and focus on route navigation and real time tracking. They fail to provide an adaptive, unified and a solution that caters to a large variety of needs by various commuters. Due to these reasons users are forced to use multiple applications and this can cause difficulty and confusion among elderly people, tourists and differently abled users. In order to address these issues, this paper proposes Namma Transit, a novel unified public transportation app. This app has been specifically designed to help the socio cultural and infrastructural context of Tamil Nadu. This system incorporates a universal user modeling system mechanism that makes the machine learning techniques to classify users based on behavioral patterns, preferences and other attributes unlike the traditional systems. The framework is designed to be scalable and adaptive making suitable for deployment across both urban and rural areas within the region.The proposed application provides a comprehensive end-to-end transportation experience by integrating functionalities ranging from route discovery to ticket booking and real time journey monitoring within this platform in addition to the adaptive and inclusive capabilities.

Keywords: *real time data streams, traffic conditions, transport schedules, dynamic route Optimization, multilingual supports for tourists, streamlined interfaces for users*

MSEC-RS-2026-ECE-001

Intelligent Passenger Demand-Based Public Transportation System

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Abstract

In modern urban environments, efficient public transportation is essential to meet the increasing demands of commuters. This project presents an intelligent passenger demand monitoring and response system that enhances transport services using real-time crowd data. Commuters register their travel destinations through a simple interface linked to their identity, ensuring unique data collection. The system continuously tracks and updates the number of passengers heading toward specific locations, using this data as a dynamic indicator of route demand. When passenger count exceeds a predefined threshold, the system identifies high-demand routes and sends real-time alerts to transport authorities, enabling the quick deployment of additional buses. This demand-driven approach overcomes the limitations of traditional fixed scheduling by adapting to real-time conditions and managing both regular and peak-hour traffic efficiently. The proposed system improves passenger convenience by reducing waiting times and preventing overcrowding, while supporting smarter decision-making in transportation management. Future enhancements such as GPS-based tracking, cloud analytics, digital ticketing, and AI-based prediction can further strengthen the system. Overall, this solution contributes to the development of intelligent, responsive, and sustainable smart city transportation infrastructure.

Keywords: *Intelligent Transportation System (ITS); Real-Time Data Analytics; Demand Monitoring; Dynamic Bus Scheduling; Machine Learning; Smart City Infrastructure*

MSEC-RS-2026-ECE-002

Intelligent Multi-Disease Diagnostic Engine

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Abstract

The use of smart monitoring systems that integrate Artificial Intelligence, the Internet of Things and embedded devices is becoming an important trend in the healthcare sector as a way of improving the care provided to patients. This project explains how a real-time physiological monitoring system that can analyze both ECG signals and respiratory lung sounds in real time can be designed and implemented to detect clinically relevant medical conditions. The hardware design focuses on a single ESP32 microcontroller that collects the data of two peripheral sensors: a heart activity sensor (ECG) and a microphone (MAX4466) sensor that record the sounds of breathing. Physiological measurements are sent through the microcontroller to a host computer via serial communication where all processing and inference is carried out. A Convolutional Neural Network that was trained on the MIT-BIH Arrhythmia Dataset is used to classify cardiac rhythms allowing to identify arrhythmia, tachycardia, and bradycardia. The pipeline of lung sound analysis consists of two steps: Mel Frequency Cepstral Coefficient (MFCC) feature extraction is used as input to a deep learning classifier that differentiates between four respiratory types: Normal, Crackle, Wheeze, and Crackle-Wheeze. Clinicians and researchers interact with the system via a web-based dashboard developed using Flask, HTML, CSS, JavaScript, and Chart.js that renders real-time ECG waveforms, lung sound traces, and diagnostic predictions. The platform can operate in two active modes; a live monitoring mode that allows observing patients in real-time, and an offline validation mode that allows controlled laboratory testing. The results of the evaluation indicate the high diagnostic accuracy as the ECG classifier had 95.3% accuracy, and the lung sound classifier had 90.2%, which proves the reliability of the system. Its portability and small size make it suitable in the context of telemedicine and remote patient monitoring.

Keywords: *Multi Disease Diagnostic; Internet of Things (IoT); Machine Learning*

MSEC-RS-2026-ECE-003

Audio-Visual Speech Enhancement and Noise Reduction

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Abstract

Speech communication systems frequently function in environments where acoustic echo and ambient noise reduce clarity and intelligibility. Traditional noise reduction techniques depend only on audio signals, which restricts their performance under difficult acoustic conditions. This work introduces an audio-visual fusion system that integrates active speaker detection, echo suppression, and noise reduction. The approach determines the speaking individual by analysing lip movements extracted from video frames through facial landmark tracking, using these motion cues to guide the overall processing. An adaptive echo cancellation stage employing the Normalized Least Mean Squares (NLMS) algorithm is used to eliminate acoustic echo from the captured microphone signal. The resulting signal is subsequently processed through spectral analysis, framing, and median filtering to facilitate accurate noise estimation. A computationally efficient speaker separation mechanism reduces interference from background noise and inactive speakers. Furthermore, a GRU-based RNN model predicts a gain mask that selectively suppresses noise while maintaining speech quality. During periods of speech inactivity, enhanced suppression is applied to avoid amplifying residual noise. Experimental evaluations indicate notable improvements in SNR, PESQ, STOI, and ESTOI under diverse noise conditions.

Keywords: *Audio-visual fusion; active speaker detection; lip movement analysis; facial landmark tracking; echo cancellation; NLMS algorithm; GRU-RNN*

MSEC-RS-2026-ECE-004

Design and Implementation of Flexible Implantable antenna for biomedical Applications

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Abstract

Implantable antennas play a crucial role in advanced biomedical telemetry systems, including pacemakers, neural stimulators, and wireless health monitoring devices. These antennas are required to function efficiently within biological tissues that tend to absorb electromagnetic signals, while also maintaining a compact form factor and ensuring safe levels of electromagnetic exposure. This work presents the design and analysis of a compact implantable antenna operating in the 2.45 GHz ISM band using CST Microwave Studio. The proposed antenna employs a miniaturized planar multilayer structure, integrating a shorting element along with a circular slot configuration to achieve size reduction. Performance evaluation is carried out using four biocompatible substrate materials: Rogers RT5880, Polyimide, Liquid Crystal Polymer, and Polydimethylsiloxane. Simulation results indicate resonance near 2.45 GHz, with a minimum return loss of −14 dB and a bandwidth reaching up to 500 MHz depending on the substrate used. The antenna exhibits a far-field directivity of 5.27 dBi, while the radiation efficiency is approximately −33.63 dB due to absorption losses in biological tissues, which aligns with reported results in implantable antenna studies. Among the materials considered, polyimide provides the widest bandwidth and is therefore identified as the most suitable choice for flexible implantable antenna applications. Overall, the results highlight that substrate selection, miniaturization techniques, and interaction with biological tissues collectively influence the performance of implantable antennas.

Keywords: *Implantable antenna; biomedical telemetry; 2.45 GHz ISM band; CST Microwave Studio; multilayer planar antenna; Rogers RT5880*

MSEC-RS-2026-ECE-005

Design and Modeling of a Dual-Nanogap Triple-Heterojunction Split-Gate T-Shaped Channel Vertical DMDG-TFET with Source Pockets for Biosensing

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Abstract

This work presents a detailed sensitivity analysis of label-free FET-biosensors by comparing traditional tunnel-FETs and split-gate (SG) T-shaped channel structures with the newly proposed dual-pocket Vertical Dielectrically Modulated Double Gate (V-DMDG) TFET-biosensor. For the first time, a T-shaped channel combined with a triple heterojunction is integrated into a V-DMDG-TFET biosensor. The sensitivity analysis, carried out using Silvaco TCAD, considers both positively and negatively charged biomolecules (N bio=0) over a wide range of permittivity (K= 1 to K= 12). For specific use in Bacteriophage Gelatin (K= 12), Keratin (K= 8), (K= 6.3), and APTES (K= 3.75) sensitive detection of biomolecules specifically for biosensors was performed by etching two cavities on both gate to drain and gate to source in a sacrificial etching process at both junctions. Performance characteristics of devices investigated thoroughly based on their Threshold Voltages, Sub Threshold Slope (SS), ION current, and IOFF current. The results revealed that owing to its vertical & lateral current conduction properties in the V-DMDG-TFET biosensor, the sensitivity is enhanced by approximately 104 times more in sensitivity compared to conventional TFET biosensors. Furthermore, by analyzing the impact of steric obstruction and non-uniform receptor/probe distribution, the results investigate non-ideal sensor behaviour. Sensitivity under various partial nanogap filling conditions was evaluated. The proposed biosensor shows up to a 100-fold increase in sensitivity when compared to previously reported sensors.

Keywords: *Bio sensor, TFET, TCAD, T shaped structure*

MSEC-RS-2026-ECE-006

ECG-MSEE-Net

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Abstract

Accurate classification of electrocardiogram (ECG) beats remains a major challenge due to noise, inter-patient variability, and spectral distortions that degrade signal quality, particularly in real-time monitoring and embedded system scenarios. To address these limitations, this paper proposes ECG-MSEE-Net, a novel deep learning model for ECG beat classification that integrates a convolutional neural network (CNN), an adaptive spectral weighting mechanism based on Maximum Spectral Energy Entropy (MSEE), and a Transformer encoder. The CNN extracts discriminative local morphological features, while the MSEE-based module adaptively weights ECG segments according to their spectral quality, enhancing robustness against noisy or atypical signals. Additionally, the Transformer encoder captures long-term temporal dependencies to further optimize classification performance. The proposed model is evaluated on the MIT-BIH Arrhythmia dataset using a patient-independent protocol. Experimental results demonstrate an accuracy of 98.74%, an F1-score of 92.19%, and strong generalization capabilities, even in the presence of noise. Comparisons with state-of-the-art methods confirm that ECG-MSEE-Net effectively handles spectral anomalies and achieves competitive performance. Furthermore, validation on an embedded platform (Raspberry Pi) highlights its potential for real-time ECG monitoring and connected healthcare applications.

Keywords: *ECG; CNN; MSEE; Transformer encoder*

MSEC-RS-2026-ECE-007

Based Real Time Accient Detection and Automated Reporting System using Deep Learning

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Abstract

Road traffic accidents continue to impose significant social and economic costs worldwide. While delayed emergency response remains a concern, a major challenge in non-fatal incidents is the lack of immediate, structured, and verifiable evidence for claim processing and dispute resolution. This paper presents an AI–based real-time accident detection and automated digital documentation and reporting system using deep learning. Leveraging existing CCTV surveillance infrastructure, a Convolutional Neural Network (CNN) is trained to classify accident and non-accident scenarios from image data. Upon detecting an accident with high confidence, the system automatically generates a structured, timestamped PDF report containing visual evidence. A Fast API-based backend manages inference, report generation, while a React-based dashboard enables efficient search, retrieval, and analytics of historical incidents. Experimental evaluation demonstrates high classification accuracy, low processing latency, and real-time feasibility. Beyond emergency response support, the proposed system serves as a digital evidence generation framework for insurance claim validation and fraud mitigation.

Keywords: *Accident Detection; Deep Learning; CNN; CCTV Surveillance; Automated Reporting; Smart Cities*

MSEC-RS-2026-ECE-008

Fruit Sense Embedded Multi-Sensor Ripeness Detection System

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Abstract

harvest loss. The grading, sorting, and inspection of the fruits are not done efficiently. The quality inspection method is time-consuming. This results in reduced fruit value and increased wastage. In this paper, an intelligent fruit ripeness detection system using computer vision and deep learning is presented. In the proposed system, the fruit ripeness detection is done by using the YOLO object detection technique. Using the YOLO technique, various fruits like bananas, apples, oranges, and mangoes are classified into various ripeness levels like unripe, ripe, overripe, and rotten. In the proposed system, images of the fruits are captured by the camera module. Based on the classification results, the fruits are segregated by the control mechanism. In the experimental results section, the real-time performance of the system is validated. Moreover, the classification accuracy of the system is validated. The proposed system is non-destructive, cost-effective, and suitable for small and medium-scale applications. In this system, the fruit ripeness is detected by using the YOLO object detection technique. The proposed system is useful for the automation of the fruit supply chain.

Keywords: *Multi sensor fusion; YOLO*, *Machine Learning; Internet of Things (IoT)*

MSEC-RS-2026-ECE-009

Intelligent Multipurpose Robotic Rover for Healthcare Assistance and Monitoring

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Abstract

In response to current environmental and health challenges, this project presents an AI-powered multipurpose rover designed for intelligent condition monitoring using the Raspberry Pi 4 Model B. The system integrates multiple sensors, including air quality, temperature and humidity, body temperature, heart rate, light intensity, and a capacitive rain sensor, to continuously collect real-time data related to environmental and human health conditions. The acquired data is transmitted wirelessly via a Flask API over a Wi-Fi network for further processing and AI-based decision-making. Machine learning algorithms such as Support Vector Machines and Linear Regression are utilized due to their high accuracy in classifying environmental conditions as safe or unsafe and in determining whether an individual′s health status is stable or critical. The rover′s navigation and motion control are managed by a separate microcontroller, the ESP8266 board, ensuring efficient and flexible operation. Additionally, the system is equipped with an alert mechanism that sends notifications to concerned individuals through WhatsApp using the Twilio API, enabling timely intervention when abnormal conditions are detected. The rover features a robust four-wheel design with cone-shaped supports to enhance stability and performance across varied terrains. For validation and justification, the proposed system has been tested using ROS simulation. Overall, this solution offers a smart, reliable, and cost-effective approach for automated monitoring in both environmental and healthcare domains, achieving a classification accuracy of 100% for environmental conditions and 98.33% for human health data.

Keywords: *AI-powered Rover; Environmental Monitoring; Health Condition Monitoring; Machine Learning; Internet of Things (IoT)*

MSEC-RS-2026-ECE-010

Vision-Verified Autonomous Retail Trolley with Secure Edge Billing Architecture

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Abstract

In recent years, supermarkets and retail stores are facing problems such as long waiting time at billing counters, billing errors, and increased dependency on manual labor. Customers often experience inconvenience due to standing in long queues, especially during peak hours. To overcome these problems, this project proposes a Vision-Verified Autonomous Retail Trolley with Secure Edge Billing Architecture that enables automatic product verification and billing directly within the trolley. The proposed system uses a barcode scanner to identify products selected by the customer during shopping. After scanning, a load cell sensor verifies the weight of the product to ensure that the correct item has been placed inside the trolley. In addition, a camera module is used to confirm the presence of the product, which improves the accuracy and security of the billing process. This multi-layer verification approach helps reduce billing errors and prevents unauthorized product placement. Unlike traditional smart billing systems that depend on internet connectivity, the proposed system performs all operations locally using an embedded controller. This edge-based processing improves system response time and allows the trolley to function even without network connection. It also improves data security by reducing dependency on external servers. The trolley follows the user in real time using Raspberry Pi Pico-based motor control modules, ensuring smooth and flexible movement inside the store. This approach eliminates the need for predefined paths and allows the trolley to move freely with the customer. As a result, it reduces physical effort and improves overall shopping convenience.

Keywords: *Autonomous Retail Trolley; Edge Computing; Smart Billing System; Barcode Scanner; Load Cell Sensor; Computer Vision; Embedded Systems*

MSEC-RS-2026-ECE-011

A Decentralized Low-Power Lora Mesh System for Infrastructure Independent Disaster Communication

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Abstract

Communication infrastructure failure is one of the most critical challenges during natural disasters such as earthquakes, floods, and cyclones. Conventional cellular and internet-based communication systems often become unavailable, severely limiting coordination between affected communities and rescue teams. To address this challenge, this paper proposes LoRaMesh-Bridge, an internet-free, low-latency LoRa mesh-to-mesh disaster alert system designed for resilient emergency communication. The proposed system utilizes ESP32-based nodes integrated with LoRa transceivers to establish decentralized, multi-hop mesh networks without relying on centralized gateways. Unlike traditional LoRaWAN star topologies, the architecture enables peer-to-peer packet forwarding to improve fault tolerance and coverage. A bridge node mechanism is introduced to interconnect independent LoRa mesh clusters, enabling scalable communication across geographically separated disaster zones. Each node also hosts a local Wi-Fi access point with a captive portal interface that displays real-time disaster alerts on smartphones without requiring internet connectivity. Sensor-triggered alert generation and lightweight routing are implemented to ensure reliable and energy-efficient message dissemination. Experimental validation demonstrates stable packet transmission and low latency suitable for emergency alerts under infrastructure-less conditions. The proposed framework contributes a scalable mesh-to-mesh bridging mechanism for decentralized disaster communication, addressing limitations of existing gateway-dependent systems. The design provides a practical pathway toward resilient, infrastructure-independent emergency communication networks.

Keywords: *LoraWAN; IOT; ESP32*

MSEC-RS-2026-ECE-012

Multi-Modal Communication Device Counter using ESP32

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Abstract

Counting wireless-enabled devices within controlled physical spaces is increasingly important for security enforcement, occupancy management, and anomaly detection. Existing single-protocol solutions either rely on network infrastructure cooperation or capture personally identifiable information, limiting their practical deployment. This paper introduces a three-modality device enumeration framework that simultaneously processes IEEE 802.11 probe requests, Bluetooth Low Energy (BLE) advertisement frames, and broadband radio-frequency (RF) envelope signals on a single ESP32 module [16]. Privacy is embedded by design: the system discards all MAC addresses and device identifiers before any storage or analysis occurs [18]. Temporal burst segmentation partitions the continuous observation stream into per-device transmission episodes, and DBSCAN clustering [11] assigns episodes to distinct device clusters using only signal-level features. Experimental cluster distributions for both WiFi and BLE modalities confirm consistent device separation across three identified clusters. The platform is assembled entirely from off-the-shelf components with low cost, making it suitable for large-scale deployment in corporate facilities, examination halls, and other restricted zones.

Keywords: *Device enumeration; privacy-preserving sensing; ESP32; WiFi probe monitoring; RF envelope detection; DBSCAN*

MSEC-RS-2026-ECE-013

Vision Assist: An Intelligent Multimodal Wearable System for Autonomous Navigation and Environmental Understanding in Visually Impaired Individuals

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Abstract

Visual impairment presents substantial challenges to independent mobility, environmental awareness, and social participation. While conventional assistive technologies such as white canes and guide dogs provide limited contextual information, emerging AI-based solutions often suffer from high deployment costs, limited personalization, and insufficient real- time processing capabilities. This paper presents Vision Assist, a comprehensive multimodal wearable vision assistance system de- signed to provide visually impaired individuals with autonomous navigation capabilities and rich environmental understanding. The system integrates a hat-mounted camera module with a Raspberry Pi 4 Model B processor, combining advanced computer vision techniques, large vision-language models (LVLMs), and haptic feedback mechanisms. Vision Assist incorporates five primary operational modes: real-time object detection with personalized recognition databases, contextual scene understanding using LVLMs, distance-based collision avoidance, face recognition with customizable user profiles, and spoken output through text-to-speech synthesis. Comprehensive user testing demonstrates significant improvements in navigation confidence, environmental awareness, and overall autonomy compared to traditional mobility aids. The proposed system is implemented using open-source technologies (AOSP, TensorFlow Lite, OpenCV) to ensure accessibility and cost-effectiveness. VisionAssist achieves a 94.2% accuracy rate in object detection and processes visual frames at 15–20 FPS on resource-constrained hardware, representing a practical solution for deployment in diverse real-world environments.

Keywords: *Computer vision; assistive technology; object detection; vision-language models; wearable devices; edge computing*

MSEC-RS-2026-ECE-014

A Dual-Band Fabry-Perot Antenna Using Stacked Patches and Spatially Separated Superstrate for CubeSat Communication

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Abstract

This paper presents the design, development, and simulation of a reconfigurable dual-band circularly polarized (CP) CubeSat antenna for C-band and X-band communication applications. The proposed antenna is designed using a multi-layer stacked patch configuration on Rogers RT/Duroid 5880 substrate (relative permittivity = 2.2, loss tangent = 0.0009). The architecture incorporates a slot-loaded stacked patch radiating element tuned independently at 6.5 GHz (C-band) for Telemetry, Tracking, and Command (TTC) operations and at 8.5 GHz (X-band) for high-data-rate payload downlink, symmetrically integrated with a positive-phase-gradient reconfigurable metasurface superstrate to achieve simultaneous gain enhancement and beam reconfigurability across both operating bands. An equivalent Fabry-Perot Cavity (FPC) resonance model based on the Spatially Separated Superstrate Area (SSSA) excitation concept is utilized to explain the independent polarization control mechanism and the role of the metasurface unit cells, PIN diode switches, and truncated-corner patch geometry in achieving wideband circular polarizationThe antenna achieves a -10 dB impedance bandwidth covering 6.2 GHz to 6.8 GHz at C-band and 8.1 GHz to 8.9 GHz at X-band, with a realized gain of 6 dBi and 10 dBi respectively, an axial ratio below 3 dB at both operational frequencies, and a radiation efficiency exceeding 85%. A Flexible Printed Circuit (FPC) based deployable mechanism enables compact stowage during launch and full operational deployment during mission.

Keywords: *CubeSat Antenna; Dual-Band; Circular Polarization; Fabry-Perot Cavity; Metasurface; Reconfigurable Antenna*

MSEC-RS-2026-ECE-015

Deep Learning Based Epileptic Seizure Detection Using EEG Signals

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Abstract

Epilepsy is a neurological disorder characterized by sudden and recurrent seizures caused by abnormal brain activity. Electroencephalography (EEG) signals are widely used for detecting epileptic seizures. However, manual analysis of EEG signals by neurologists is time consuming and prone to human error. This paper proposes a deep learning based epileptic seizure detection system using EEG signals. The proposed model combines Convolutional Neural Networks (CNN) and Long Short Term Memory (LSTM) networks to automatically extract spatial and temporal features from EEG signals. This project proposes an automated seizure detection system using EEG signals obtained through Electroencephalography. Experimental results demonstrate that the proposed system achieves an accuracy of approximately 98outperforming several traditional machine learning approaches. The proposed method can assist neurologists in faster and more reliable seizure detection. Experimental results show that the proposed CNN–LSTM model can effectively learn EEG patterns and achieve high accuracy in seizure detection. The developed system demonstrates the potential of deep learning techniques in assisting clinicians for automated and real-time monitoring of epileptic seizures.

Keywords: *EEG; Epileptic Seizure Detection; Deep Learning; CNN; LSTM; Biomedical Signal Processing*

MSEC-RS-2026-ECE-016

Image Super Resolution for Morphological Studies for Tumor Cells using DL

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Abstract

Cancer cell morphology analysis plays a critical role in understanding disease progression, tumor states, and therapeutic responses. However, acquiring high-resolution microscopy images remains challenging, as it typically requires expensive imaging systems and specialized optical techniques. In this work, we propose a computational super-resolution framework to enhance the resolution of cancer cell microscopy images using a modified Deep Fourier Channel Attention Network (DFCAN). The primary objective is to improve the visualization of fine cellular structures from low-resolution inputs. By leveraging frequency-domain feature representation and channel attention mechanisms, the proposed model effectively enhances high-frequency structural details while preserving overall cellular morphology. The network was trained on a GPU-based computing platform with a total training time of approximately 16 hours. Experimental results demonstrate that the modified DFCAN achieves strong performance, with a peak signal-to-noise ratio (PSNR) of 41.5 dB and a structural similarity index (SSIM) of 0.98, indicating high fidelity between reconstructed and ground-truth images. The super-resolved images provide clearer visualization of cancer cell morphology, including well-defined structural boundaries and intracellular features that are essential for biological interpretation. This approach offers a cost-effective, non-invasive, and accessible alternative to conventional high-resolution imaging techniques.

Keywords: *Image Super-Resolution; Cancer Cell Morphology; Deep Learning, DFCAN (Deep Fourier Channel Attention Network; Microscopy Image Enhancement; Frequency Domain Analysis; Structural Similarity (SSIM)*

MSEC-RS-2026-ECE-017

Explainable Lung Disease Classification from Chest X-Ray Images Using CLAHE and Swin Transformer

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Abstract

Lung disease classification using chest X-ray (CXR) images is a critical task for early diagnosis and treatment planning. Although existing deep learning models have demonstrated promising performance, they often suffer from limitations such as inadequate feature extraction, lack of region-focused learning, and poor handling of class imbalance. In this work, an improved hybrid framework is proposed by integrating preprocessing, segmentation using Otsu thresholding, data augmentation, and a Swin Transformer-based architecture for feature extraction. A hybrid loss function combining weighted cross-entropy and focal loss is employed to enhance classification performance, particularly for difficult samples. The model is evaluated on a multi-class lung disease dataset consisting of COVID-19, Normal, and Pneumonia images. Experimental results demonstrate that the proposed method achieves an accuracy of 96.2%, outperforming the existing approach. The improved performance is attributed to effective region-of-interest extraction, robust feature representation, and balanced optimization. This framework shows strong potential for real-world clinical applications.

Keywords: *Lung cancer feature extraction; Deep convolutional neural network; Hybrid methodology*

MSEC-RS-2026-ECE-018

Multiple Sclerosis (MS) is an inflammatory disorder using Deep learning Method

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Abstract

Multiple Sclerosis (MS) is an inflammatory disorder of the central nervous system that is considered chronic. A person with MS may lose myelin as well as experience a variety of physical/mental symptoms. There are many types of symptoms and varying degrees of severity. There are many different types of immune system dysfunctions or problems that lead to MS and can cause damage to the nervous system and other health problems related to the immune system. Many researchers believe that MS is an immune-mediated disease, however, how MS is classified remains TBD and disputed, and many researchers will say MS is an autoimmune disease, but there is not enough conclusive evidence to classify it this way currently. There are some research studies that dispute the notion that MS is autoimmune, so there is evidence indicating that there are factors outside of autoimmunity that cause MS.

Keywords: *Multiple Sclerosis (MS); Central Nervous System (CNS); Demyelination; Immune-mediated disease; Autoimmune disease (debated)*

MSEC-RS-2026-ECE-019

Design and Simulation-Based Analysis of an Energy Efficient Ultrasonic Dishwasher Using Piezoelectric Transducers

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Abstract

Traditional dishwashers consume considerable amounts of electricity and water, making them less suitable for sustainable household use. This paper proposes the design and simulation- based evaluation of an energy – efficient ultrasonic dishwasher that operates using piezoelectric transducers at 40 kHz. The system converts a standard AC supply into a regulated high- frequency signal to drive the transducer, producing ultrasonic vibrations. These vibrations create cavitation bubbles in water, which collapse and remove dirt from utensils without physical contact. Circuit simulations were conducted to verify stable 40 kHz signal generation, proper power regulation, and frequency- domain characteristics through FFT analysis. Results demonstrate effective ultrasonic excitation with low power consumption (approximately 60 W), highlighting the system′s potential as a sustainable and efficient alternative for modern dishwashing applications.

Keywords: *Ultrasonic cleaning; piezoelectric transducer; acoustic cavitation; MATLAB simulation; PZT-4; water conservation*

MSEC-RS-2026-ECE-020

Analog and Digital Signal Monitoring Over RS-485 Using MODBUS RTU

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Abstract

This project presents the design and implementation of an industrial communication-based monitoring system titled "Analog and Digital Signal Monitoring over RS-485 using Modbus RTU." The system enables real-time supervision and reliable data transfer of critical parameters in industrial battery chargers, ensuring operational safety, fault detection, and process efficiency. A local (Slave) unit is implemented using an Arduino Uno, where analog inputs from trim potentiometers simulate voltage conditions, and a push button represents fault signals. The analog signals are digitized via the Arduino's inbuilt ADC, analyzed, and categorized according to predefined thresholds to detect abnormal operating conditions accurately. The processed data is structured using the Modbus RTU protocol and transmitted over an RS-485 interface via MAX485 transceivers, enabling the integration of multiple Slave units into a centralized monitoring framework. In the prototype, communication between a single Master and Slave was successfully demonstrated, with the Master continuously polling the Slave and displaying the received data on an LCD for real-time monitoring. Immediate local fault indications are provided through LEDs and a buzzer, complemented by an acknowledgment mechanism to enhance operator convenience and system safety. By combining reliable industrial communication, threshold-based fault detection, and operator-friendly feedback, the system provides a robust, cost-effective, and efficient framework suitable for integration into industrial battery charger systems. This approach demonstrates versatility and practical relevance, making it applicable to a wide range of industrial automation environments requiring multi-device monitoring and real-time supervision.

Keywords: *EEG; Epileptic Seizure Detection; Deep Learning*

MSEC-RS-2026-ECE-021

Smart Assistive Navigation System for Visually Impaired

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Abstract

This project presents a portable, high-performance assistive device designed to enhance the autonomy of visually impaired individuals through real-time environmental perception and hybrid intelligence. Built on a Raspberry Pi, the system leverages MediaPipe and Efficient Det-Lite to perform offline detection of 80+ object categories, ensuring user privacy and low-latency processing. By integrating an ultrasonic sensor and a multithreaded architecture, the device provides instantaneous audio feedback (e.g., ″Chair at 120 cm") at a fluid 15–20 FPS. To transcend basic obstacle avoidance, the framework introduces a Cloud-Edge Hybrid ″Expert Mode" that triggers when the system encounters complex or unknown environments. In this mode, the device offloads high-resolution imagery to powerful Large Multimodal Models (LMMs), such as Gemini Pro Vision or GPT-4o, for a deep-scene analysis. This integration enables sophisticated capabilities including environmental literacy via Optical Character Recognition (OCR), semantic scene summarization for contextual awareness, facial recognition for social inclusion, and the authentication of currency and legal documents. By synthesizing reliable local detection with on-demand generative AI, this solution offers a low-cost, scalable, and highly accurate navigation ecosystem that fosters greater independence for the visually impaired.

Keywords: *Assistive device; visually impaired navigation; Raspberry Pi; Media Pipe; Efficient Det-Lite; cloud-edge hybrid system*

MSEC-RS-2026-EEE-001

Design and Implementation of a CNC-Based Braille Embosser with Multi-Input Capability for Assistive Learning Applications for the Visually Impaired

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Abstract

Access to quality Braille education is limited by the high cost of commercial embossers and their lack of input options. This paper describes the design and implementation of a low-cost, open-source multi-input CNC Braille embosser for visually impaired. The proposed system supports the users to provide input in the form of speech, typed text, or digital documents (PDF/DOCX) via a unified browser-based interface. The system integrates the Web Speech API for real-time voice recognition, along with a custom Python-based UEB Braille translation engine. It generates G-code, which is transmitted to an Arduino UNO running GRBL 1.1 firmware. A three-axis CNC mechanism, driven by NEMA 17 stepper motors, DRV8825 drivers, and a servo-based embossing head, produces raised Braille dots on standard A4 paper within a working area of 190 × 227 mm. The system accommodates 26 cells per line and 21 lines per page, resulting in a total of 546 cells. In contrast to existing single-input hardware solutions and software-only approaches, the proposed system integrates multi-modal input, a real-time text preview for error correction, and a web-based interface that eliminates the need for prior knowledge of Braille, all at a significantly lower cost than commercial alternatives. Experimental results demonstrate reliable end-to-end embossing accuracy for the complete English alphabet, numerals, and punctuation symbols.

Keywords: *Braille embosser, CNC, GRBL firmware, G-code, Arduino, assistive technology*

MSEC-RS-2026-EEE-002

Dynamic Performance Investigation and Realization of Driven BLDC Motor for High-Thrust Propulsion Systems

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Abstract

Precise motor control and real-time monitoring are critical requirements in modern brushless DC (BLDC) motor drive systems for robotics, UAV propulsion, and embedded automation applications. This project presents the design and implementation of a custom Electronic Speed Controller (ESC) for a BLDC motor constructed from scratch, employing sensorless back-EMF zero-crossing detection for commutation control without the need for Hall-effect position sensors. The system utilizes a two-microcontroller architecture combining an ATmega328P (Arduino) for low-level PWM generation and commutation sequencing with an ESP32 for high-level monitoring, communication, and web-based control. The power stage is implemented using IRLZ44N MOSFETs in a three-phase inverter bridge configuration driven by IR2101 half-bridge gate driver ICs, operating at a nominal bus voltage of 12 V. Back-EMF zero-crossing detection is realized through a resistor voltage divider network (39 kΩ / 10 kΩ) on all three motor phases. A real-time web dashboard hosted on the ESP32 Wi-Fi hotspot enables wireless motor speed control, live RPM display, current and voltage telemetry, motor health estimation, and time-series data logging through a browser-based interface accessible without external network infrastructure. Experimental results demonstrate reliable senseless start up, stable open-loop speed regulation, and responsive web-based control.

Keywords: *Brushless DC motor, senseless ESC, back-EMF zero-crossing detection, IRLZ44N MOSFET, IR2101 gate driver, three-phase inverter bridge, ATmega328P, ESP32, Wi-Fi hotspot, real-time motor monitoring, web dashboard, PWM commutation, Ki Cad PCB design.*

MSEC-RS-2026-EEE-003

Plant Leaf Disease Detection Using IOT and Machine Learning

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Abstract:

Plant diseases are generally caused by pest insect and pathogens and decrease the productivity to large-scale if not controlled within time. This paper presents a system for detection of diseases on plant leaf along with soil quality monitoring. The work proposes a Internet of Things [IoT] based Convolutional Neural Network (CNN) Machine learning regression system for identification and classification of plant leaf diseases like Bacterial Blight, Alternaria, Gray Mildew, Cereospra, and Fusarium wilt. After disease detection, the name of a disease will be provided to the farmers using android app. The Android App is also used to display the soil parameters values such as humidity, moisture and temperature. All this leaf disease detection system and sensors for soil quality monitoring are interfaced using Raspberry Pi which make it independent and costeffective system.

Keywords: *CNN, ML, Raspberry Pi, IoT*

MSEC-RS-2026-EEE-004

Centralized Monitoring and Controlling Smart Circuit Breakers

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Abstract

This paper presents an IoT-based system for monitoring and controlling circuit breakers using an ESP32 microcontroller for realtime data processing. The system continuously monitors voltage, current, and temperature across two independent channels, namely MCB1 and MCB2. Integrated sensors ensure accurate measurement of electrical parameters under varying conditions. The system detects overvoltage conditions above 240V and undervoltage conditions below 200V, ensuring reliable operation. It also monitors temperature to identify overheating conditions above 33°C. An automated protection mechanism is implemented to trip the circuit during fault conditions, preventing equipment damage and reducing fire hazards. A 20x4 I2C LCD is used for local visualization of real-time data. In addition, the monitored data is transmitted to the Blynk IoT Cloud platform via Wi-Fi for remote access. Users can monitor system performance through a mobile application interface. The system also allows manual remote control of circuit breakers using relay modules. This improves operational flexibility and user convenience. The proposed design enhances electrical safety and system reliability. It significantly reduces downtime by enabling early fault detection and rapid response. The system is cost-effective, compact, and easy to implement in real-world scenarios. It is also scalable and suitable for residential, industrial, and smart grid applications. Experimental results demonstrate fast response time and consistent performance under different operating conditions. Overall, the system provides an efficient solution for smart electrical monitoring and control, contributing to improved safety, reliability, and energy management.

Keywords: *IoT (Internet of Things), ESP32 Microcontroller, Circuit Breaker Monitoring, Blynk IoT Cloud, Remote Control System, Electrical Safety.*

MSEC-RS-2026-EEE-005

Investigations into Emergent Phyllotactic Patterns in Multi-Robotic Solar Tracking Systems using Bi-Level Optimization to mitigate Dynamic and Partial Shading conditions in PV Solar System

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Abstract

The present work investigates the emergent spatial organization of a multi-agent solar tracking system. The work is an extension from a single-PV cell evolvable robotic arm model. In the original system, a photovoltaic cell mounted on a three-level servo motor based robotic arm employs bi-level optimization integrated with Genetic Algorithms to mitigate dynamic and partial shading conditions. The present work has multiple robotic arms each operating individually without any coordination or predefined spatial arrangement specified. The conventional bio-inspired designs work with constraints in which present work differs. The investigation focuses on whether the no-constraint optimization behaviour of individual robotic arms leads to the spontaneous emergence of structured spatial configurations. The system is, however, disturbed by dynamic and partial shading patterns similar to the single robotic arm-based system. The disturbance is augmented by the fact that in addition to the externally induced partial shading patterns to the system, each of the robotic arm shades the other during the optimization process. This interaction is key to the emergent pattern of spatial organization. The primary focus of the investigation is to study whether the emergent pattern follows the natural phyllotactic patterns followed by the plants. The system performance and structural orientation are assessed by simulations are carried out under varying irradiance patterns and dynamic and partial shading patterns. Various metrics like angular distribution, shading index, and overall energy efficiency are determined to quantify the degree of organization and effectiveness. The aim is to understand whether the natural patterns can emerge from the de-centralized, no-constraint driven optimization methods thereby ascertaining the relationship between evolutionary computation and biological organization in engineering systems. The investigation is also helpful to design scalable, efficient and reconfiguring solar energy harvesting systems under dynamic and partial shading conditions.

Keywords: Emergent Behaviour, Phyllotaxy, Phototropism, Bi-Level Optimization, Genetic Algorithm, Multi-Agent Robotic Systems

MSEC-RS-2026-EEE-006

AI-Powered Assistive Device for Visually-Impaired

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Abstract

Disabled people greatly struggle accessing information like an ordinary person. People with visual impairments often face many challenges. One of the most common difficulties is accessing information in everyday setting or notice boards and written text during lessons or meeting which can limit their participation and inclusivity. While technology has advanced in providing assistive tools, most solutions rely on internet connectivity, specialized equipment and often comes with high cost.

This project aims to provide a portable, standalone device that assist the user in everyday life as a supportive device. Where user could interact and get answers with just voice. The system processes the information locally, and generate the output and give it as audio-based output to the user. By combining accessibility, portability and offline functionality, the designed system seeks to enhance learning opportunities and promote the inclusivity

Keywords: *Assistive Device, Raspberry Pi, Ollama, Edge AI, Optical Character Recognition (OCR), Image Processing, TTS and STT, Micro-controller*

MSEC-RS-2026-EEE-007

AI-Driven Battery Health Prediction and Soft-Switching Converter Optimization for Electric Vehicles

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Abstract

This study presents a novel photovoltaic (PV)-assisted electric vehicle (EV) system that integrates renewable energy conversion, intelligent energy management and predictive battery health monitoring to enhance overall performance and sustainability. The proposed system provides a unique contribution combining solar power generation, battery storage and an Active Controlled Smart Switching (ACSS) bidirectional converter with Deep Neural Network (DNN) to ensure efficient energy transfer between the battery unit and Brushless DC (BLDC) motor. A three-phase Voltage Source Inverter (VSI) supports smooth motor operation, while Proportional–Integral (PI) controllers optimize voltage, current and speed regulation under dynamic conditions. The DNN- based Battery Monitoring System (BMS) is capable of accurately estimating State of Charge (SOC) and State of Health (SOH), thereby enabling predictive maintenance and extending battery lifespan. System's functionality is confirmed using MATLAB simulations and hardware experiments under varying temperature, irradiance and PV shutdown conditions. Results demonstrate stable power exchange, improved energy utilization and a peak efficiency of 96.5 %, confirming the effectiveness of the proposed converter and intelligent BMS integration for next-generation, high-performance EV applications.

Keywords: *PV system, ACSS converter, DNN, BMS, VSI, BLDC motor, EV*

MSEC-RS-2026-EEE-008

A Novel Intelligent System for Automated Aquarium Environment Monitoring and Control

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Abstract

Maintaining a healthy aquarium requires regular monitoring of important water parameters such as temperature, pH level, turbidity, and water level. These factors directly affect the health and survival of aquatic organisms. In traditional systems, aquarium maintenance is done manually, which needs continuous attention and can lead to delays in identifying harmful changes in water conditions. Such delays may cause stress to fish and degrade water quality.

This paper presents a novel intelligent system designed to automatically monitor and control aquarium conditions using sensor-based and embedded technologies. The system uses multiple sensors to continuously measure key environmental parameters and send real-time data to a microcontroller. The collected data is compared with predefined safe limits to ensure proper habitat conditions.

When any parameter goes beyond the safe range, the system automatically performs corrective actions such as activating cooling devices, filtration units, water pumps, or other control mechanisms. In addition, an automated feeding system is included to provide food to the fish at scheduled intervals, ensuring consistent and proper feeding.

The proposed system reduces the need for manual intervention and provides continuous monitoring of the aquarium environment. By taking immediate corrective actions, it helps maintain a stable and healthy habitat for aquatic life while minimizing the effort required from the user.

Keywords: *Aquarium Monitoring, IoT, Embedded Systems, Water Quality Sensors, Automated Control, Real-Time Monitoring, Smart Aquarium, Aquatic Environment*

MSEC-RS-2026-EEE-009

Precision Agriculture: Spotting and Spraying Yellow Defected Plant

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Abstract

The agricultural sector plays an important role in the process of economic development of a country. The conventional pesticide spraying method involves uniform spraying of the entire field even when only some plants are infected. This results in pesticide wastage, increased operation costs, and health hazards. Hence, there is a strong need for an automated, precise, and selective spraying of the plants. This project presents a Precision Agriculture System using Drone Technology for Spotting and Spraying Yellow Defected Plants. The proposed system will consist of a quadcopter with a high-resolution camera and an image processing tool. The captured image is processed and compared with various examples, and the suitable control signal is given to the spraying mechanism for the spraying of the defected plant. This selective spraying reduces the waste of pesticides and decreases the environmental impact while ensuring proper care of the plants. This system integrates the Robot Operating System (ROS) for the control and communication, while the OpenCV enables the real time image processing of the defective plant. The drone mechanical build includes a light weight carbon fibre body with high resolution camera and 3D printed custom parts for delicate electronics. Together these components ensure accurate detection and spraying of the defected plants. The proposed system offers several advantages such as reduced zero chemical usage, improved crop yield, lower labour requirements, and enhanced farming efficiency. The future work will focus on the integration of the advanced machine learning models to accurately find the defected plant and enhancing autonomous path planning through ROS for optimized field coverage. This project contributes toward the development of sustainable and intelligent farming solutions in modern precision agriculture.

Keywords: *Agriculture, Drone, ROS, OpenCV, Fusion 360*

MSEC-RS-2026-EEE-010

Ambulance Priority Traffic Management System

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Abstract

Traffic congestion in cities frequently causes delay to emergency vehicles such as ambulances. This can be the difference between life and death when emergencies happen. This project offers an efficient way to get ambulance to accidents quickly by using an intelligent Ambulance Priority Traffic Management System. Based on embedded technology, the real time detection approaches can be used to determine the arrival of an ambulance before a signal. Approaches are various like wireless transceiver modules (RF, GSM, or IoT based systems), sound detectors (detect siren), GPS, etc. As soon as an ambulance is known to reach near a junction, the traffic light sequence is stopped and the close lane signal is set to green for the damage free passage of the ambulance. Once the ambulance has traveled beyond the signal and has reached the hospital region the system intelligently puts the signals back into their normal sequence of operation. This happen with least disturbance to the normal signal flow. The suggested system is realized with the use of a microcontroller-based system (e.g. Arduino, STM32), coupled with sensors and communication modules. The system improves safety on the roads, speeds up the speed of response of the Ambulances, and supports the infrastructure of smart cities.

In general, the project results an essentially efficient solution for emergency traffic management by providing a rapid helping for patient.

Keywords: *Ambulance Priority, Traffic Signal Control, Emergency Vehicle Detection, SmartTraffic Management, Traffic Congestion Reduction, Microcontroller (STM32) Real-Time Monitoring, Embedded System*

MSEC-RS-2026-EEE-011

Harmonic Mitigation and Total Harmonic Distortion Analysis in Grid Connected Voltage Source Converter Systems

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Abstract

Power electronic converters are really important for electrical power systems because they help convert DC power to AC power efficiently. These converters can also cause problems because they switch on and off really fast which creates harmonic distortion in the system. This distortion can hurt the power quality make the system lose power and even cause sensitive equipment to malfunction. This study looks at how to reduce distortion and analyze Total Harmonic Distortion in a grid-connected three-phase Voltage Source Converter system. Sinusoidal Pulse Width Modulation is used to create switching pulses for the inverter, which helps control the output voltage and reduces lower-order harmonics. An LCL filter is used to the output to suppress high-frequency switching harmonics and improve the waveform quality. A closed-loop control strategy based on Park transformation is used to make the system work better. This transformation helps simplify control and allows us to regulate reactive power components independently. Integral controllers with decoupling is used to get precise current control and improve the dynamic response.The entire system is simulated using MATLAB to see how it works under different conditions. Fast Fourier Transform analysis is used to determine the spectrum and quantify Total Harmonic Distortion levels below 2% when we use the LCL filter and closed-loop. The results show that our approach reduces Total Harmonic Distortion from around 18-20% to control together.The results confirm that the system improves power quality produces, near-sinusoidal output waveforms and ensures operation of grid-connected converter systems.

Keywords: *Harmonic mitigation, Total Harmonic Distortion, Voltage Source Converter, LCL filter, Park transformation, dq control, SPWM, grid-connected inverter.*

MSEC-RS-2026-EEE-012

Real-Time Human Posture Recognition System for Ergonomic Monitoring Using Machine Learning

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Abstract

This project describes the development of an Intelligent Posture Monitoring System (ISPMS) that can identify different sitting postures in real time using a minimal setup of four Force-Sensitive Resistors (FSR-406) placed within a chair cushion. The pressure data from these sensors is processed through voltage divider circuits, converted into digital signals using dual ADS1115 16-bit analog-to-digital converters, and transmitted to a computer via an ESP32-S3 microcontroller at a rate of one sample per second. A Python-based processing pipeline performs sensor-specific baseline calibration and applies asymmetric exponential moving average (EMA) filtering to smooth the data. From the processed signals, fifteen meaningful features are extracted, such as symmetry ratios between left and right sensors, center-of-pressure positions, and diagonal pressure differences. The system was trained and tested using a dataset of 2,659 labeled samples collected from six individuals, covering five common posture categories: Straight, Left Lean, Right Lean, Left-on-Right, and Right-on-Left. Among six machine learning models evaluated, XGBoost delivered the best results, achieving 99.06% accuracy on the test dataset and an average accuracy of 99.25% (±0.0034) using 5-fold cross-validation, demonstrating consistent performance. Overall, the results show that a simple and affordable four-sensor arrangement, when combined with effective feature extraction and machine learning, can reliably classify sitting postures in real time, making it a practical solution for ergonomic monitoring in office settings.

Keywords: *FSR, XGBoost, ML, EMA, ESP32-S3, IoT*

MSEC-RS-2026-EEE-013

Designing and Implementation of a Soft-Switched High-Gain DC-DC Converter for Negative Boost Applications

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Abstract

Many converters are available in DC-DC converters. The proposed converter is negative voltage boost converter. Since negative voltage has a lack of electrons, it can neutralize the positive charge that can avoid producing heat. A negative voltage rail is required for high performance automation devices. As the Negative Boost Converter's applications, span various industries. Its ability to efficiently increase or decrease voltage levels suits an attractive choice for power conversion applications requiring high efficiency, reliable operation, and bidirectional-power-flow capability. The design and implementation of soft switching in negative boost converters aims to maximize the efficiency and performance of power conversion systems. Boost converters used in various applications because of their ability to provide voltage step-up and step-down capabilities. However, they suffer from switching losses and high stress on semiconductor devices, leading to decreased efficiency and increased heat generation. This paper presents a novel soft switching technique applied to negative boost converters, which mitigates switching losses (SL) and minimizes the stress on power switches.

Keywords: Negative boost Converter, Power electronics, Soft switching(SS), Switching losses(SL), Voltage stress, Zero voltage switching (ZVS)

MSEC-RS-2026-IT-001

A Virtual Cognitive Behavioural Therapy System for Juvenile Rehabilitation Based on Becks Cognitive Model using DistilBERT

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Abstract

The growing prevalence of mental health challenges among juveniles, combined with limited access to professional counseling, highlights the need for scalable and accessible therapeutic solutions. Existing AI-based mental health systems often function as general conversational agents without a structured Cognitive Behavioral Therapy (CBT) framework, limiting their ability to detect cognitive distortions, emotional patterns, and behavioral risks. To address this gap, this project proposes a Virtual Cognitive Behavioral Therapy System for Juvenile Rehabilitation based on Beck's Cognitive Model using DistilBERT. The system integrates emotion recognition, cognitive distortion detection, and intent classification to deliver structured and personalized therapeutic support. Using natural language processing and transformer-based models, it analyzes user inputs (text or speech), identifies negative thought patterns, and generates adaptive CBT-based responses. The architecture consists of user interaction, AI-based psychological analysis, CBT reasoning, response generation, and ethical data management modules. It also includes risk detection, session memory, and supervised monitoring for safe and continuous support. Focused on early-stage mental health assistance, the system ensures accessibility, multilingual adaptability, and ethical deployment, providing scalable and personalized support for juvenile rehabilitation.

Keywords: *Cognitive Behavioral Therapy (CBT), Juvenile Rehabilitation, DistilBERT, Emotion Detection, Intent Classification, Risk Prediction*

MSEC-RS-2026-IT-002

An Interpretable Multimodal Framework for Knee Osteoarthritis Using Swin Transformer

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Abstract

Accurate early-stage knee osteoarthritis (KOA) detection from X-ray images remains challenging due to subtle radiographic changes and the limited interpretability of existing AI-based diagnostic systems. To address this limitation, this work proposes an interpretable SWIN Transformer–based AI framework for early knee osteoarthritis detection and severity grading using radiographic images and patient clinical data. The proposed approach focuses on improving early-stage KOA identification while delivering reliable and clinically interpretable diagnostic outcomes. The methodology employs a SWIN Transformer–based deep learning model for X-ray feature extraction, an MLP for clinical data processing, and a diffusion-based fusion mechanism with an explainability module to perform Kellgren–Lawrence severity grading. Experimental evaluation demonstrates improved accuracy and consistent performance in both detection and severity assessment. Overall, the proposed framework enhances diagnostic reliability and supports transparent, clinician-assisted knee osteoarthritis evaluation. Future work will evaluate the framework on multi-center datasets to improve generalization across different hospitals and imaging devices.

Keywords: *SWIN Transformer, Multimodal learning, Explainable AI, Kellgren–Lawrence grading, Clinical decision support*

MSEC-RS-2026-IT-003

An Intelligent Advisory Based Farmer Grievance System Using Deep Learning

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Abstract

The farmers in rural areas who speak multiple languages face problems because they cannot access both quick grievance resolution and dependable agricultural advisory services. Manual complaint handling systems create problems because they make it difficult to process complaints, leading to incorrect complaint classification, less important complaints not being resolved, and the system losing trust from users who depend on digital governance solutions. This study presents an AI-based Farmer Grievance and Advisory System using BERT-based natural language processing technology to handle complaints and provide advisory support in multiple languages. The system processes farmer submissions through text and voice inputs in regional languages, featuring speech-to-text conversion, language identification, translation, and structured preprocessing. The BERT classification engine applies fine-tuned analysis to determine contextual intent, analyze complaint types, assign departmental responsibility, and establish priority ranking through transformer-based semantic comprehension. The system architecture combines automated ticket creation with intelligent ticket distribution, workflow status monitoring, and AI-powered advisory response development in a secure and scalable environment. The system achieves better classification accuracy and shorter response times through multi-stage training using evenly distributed datasets, attention-based contextual modeling, and performance optimization methods.

Keywords: *Farmer Grievance System; BERT; NLP; Deep Learning; Multilingual Processing; Advisory System; AI in Agriculture*

MSEC-RS-2026-IT-004

Tolkāppiyam: An AI-Driven Ontology and Knowledge Graph for Tamil Linguistic and Cultural Knowledge

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Abstract

An AI-based knowledge modeling solution for ***Tolkāppiyam*** is presented, employing an ontological approach to knowledge modeling by systematically encoding linguistic, grammatical, and cultural knowledge contained in classical Tamil literature. The scholars extracted textual information from reliable digital sources and organized them into various categories, such as ***Ezhuthu, Sol, Porul, Akam, Puram,*** and ***Thinai,*** to indicate the relationships between concepts. The solution applies ontology engineering and knowledge graph creation to develop semantic models of grammatical entities linked to themes and cultural components. Advanced graph analysis and canonical query evaluation, multi-hop subgraph extraction, and inference techniques help researchers identify dense conceptual communities and influence relations and semantically meaningful nodes in the data. Sutra and sections identification for each concept and relation assist in tracing their origins, making the process transparent and interpretable. The solution provides a knowledge support framework integrating explainable intelligence and data capabilities for preserving classical Tamil knowledge through structured investigation and analysis. This groundbreaking design facilitates the integration of semantic technologies and computational humanities research for improved structured knowledge discovery, maintaining cultural heritage and interpretive capabilities while advancing digital academic studies.

Keywords: *Tolkāppiyam*; *Tamil Linguistics; Ontology Engineering; Knowledge Graph; Semantic Modeling; Cognitive Modeling; Explainable AI; Computational Linguistics; Cultural Knowledge Representation*

MSEC-RS-2026-IT-005

Multimodal Feature Fusion Using ResNet and Wav2vec2 for Early Neurological Risk Assessment

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Abstract

Early identification of neurological disorders remains a significant challenge due to the variability in early motor and speech-related symptoms, as well as the reliance on costly clinical diagnostic methods. This study aims to develop a multimodal deep learning framework for early prediction of neurological risk by analyzing both handwriting and vocal signals. The proposed approach employs a ResNet-based convolutional neural network (CNN) to extract spatial motor features from handwriting images, along with a Wav2Vec2-based transformer model to derive contextual acoustic representations from vocal data. These extracted features are integrated to compute the Neural Stability Score (NSS), a non-invasive and cost-effective measure that reflects overall motor and vocal stability. Experimental results demonstrate that combining multimodal features improves stability and robustness compared to single-modality methods. Overall, the framework provides an effective digital solution for early detection and prediction of neurological risks.

Keywords: *Early Neurological Disorder Detection, Multimodal Deep Learning Framework, Handwriting Analysis, Vocal Signal Analysis, ResNet-based CNN*

MSEC-RS-2026-IT-006

An Explainable Forward Chaining-Based Expert System for Women Safety and Legal Guidance

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Abstract

Women safety and access to timely legal guidance remain critical challenges due to the lack of context-aware and explainable decision-support systems. Existing digital platforms primarily provide general legal information and rely on manual consultation, which often leads to confusion, delayed response, and limited awareness of applicable legal provisions. This paper proposes an Explainable Forward Chaining-Based Expert System for Women Safety and Legal Guidance, designed to provide structured, situation- specific legal recommendations. The system collects incident-related inputs such as type of issue, location, age group, and description, and processes them using a rule-based legal knowledge base. By applying forward chaining inference, the system identifies relevant IPC sections, evaluates severity levels, and generates clear safety recommendations along with step-by-step reporting procedures and helpline information. The proposed framework enhances transparency, interpretability, and accessibility by providing explainable reasoning for each decision. It reduces dependency on manual legal consultation and improves decision-making efficiency in critical situations. Experimental results demonstrate effective incident classification, reliable severity assessment, and context-aware legal guidance generation, making the system a practical tool for improving women safety awareness and support.

Keywords: *Women Safety, Legal Expert System, Forward Chaining, Explainable AI, Rule-Based Reasoning, Legal Decision Support, Incident Analysis, IPC Sections*

MSEC-RS-2026-IT-007

AI-Based Multimodal Risk Analysis and Pattern Detection Using Machine Learning Techniques for Anonymous Safety Reporting Systems

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Abstract

Women safety remains a critical concern as many incidents go unreported due to lack of secure and anonymous communication platforms, while existing solutions are limited to reactive SOS-based approaches. This study proposes an AI-driven multi- modal framework for enabling anonymous reporting and intelligent safety analysis. The system analyzes user-generated messages using Natural Language Processing (NLP) and integrates location-based insights to perform risk scoring and pattern detection. By combining textual and spatial data, the framework supports real-time risk assessment and proactive safety measures. The proposed approach enables a scalable and intelligent system for enhancing women safety and supporting timely intervention.

Keywords: *Women Safety, Anonymous Reporting, NLP, Risk Scoring, Pattern Detection, Encryption, AES, RSA, Real-Time Alerts, Safety Systems, AI-Based Analysis*

MSEC-RS-2026-IT-008

Context-Aware Traffic Signal Recognition for CVD Drivers Using YOLOv12 & Lane Segmentation

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Abstract

Drivers with Color Vision Deficiency (CVD) face significant safety risks at urban intersections due to difficulties in distinguishing traffic signal colors. While standard deep learning models recognize colors, they often lack the contextual awareness required for complex multi-lane environments. This work introduces an assistive prototype that enhances detection through lane specific targeting and real-time distance estimation. Utilizing a YOLOv12 architecture for high-precision signal classification, the system integrates a computer vision-based lane detection module to filter out non-ego signals, ensuring alerts apply only to the driver's current path. Furthermore, a monocular distance estimation algorithm provides critical proximity warnings as the vehicle approaches intersections. Delivered via offline audio feedback in Tamil and English, the system was evaluated on a diverse dataset including night, rain, and heavy traffic conditions, achieving a (mAP) of 0.81. This intelligent assistant empowers CVD drivers with the spatial and color-based information necessary for safe and independent navigation.

Keywords: *CVD Drivers, Traffic Signal Recognition, YOLOv12, Lane Detection, Distance Estimation, Driver Assistance.*

MSEC-RS-2026-IT-009

A Random Forest Approach for Dynasty Classification of Ancient Coins Using Structured Metadata

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Abstract

A Random Forest Approach for Dynasty Classification of Ancient Coins Using Structured Metadata addresses the limitations of traditional coin identification methods that rely on manual expert analysis, which is time-consuming and difficult to scale for large archaeological datasets. Existing approaches mainly focus on image-based recognition techniques that depend heavily on visual features and often ignore structured coin information. To overcome these limitations, the proposed system develops a machine learning-based classification using Random Forest that utilizes coin metadata such as metal type, region of discovery, weight, and inscription details to predict the dynasty of ancient coins. The system improves classification accuracy and efficiency while reducing manual effort, thereby supporting digital numismatics research and cultural heritage preservation.

Keywords: *Random Forest, Machine Learning, Dynasty Classification, Ancient Coins, Structured Metadata, Coin Identification, Numismatics, Archaeological Data, Cultural Heritage, Metadata Analysis*

MSEC-RS-2026-IT-010

Attribute based approaches for secure data sharing in industrial contexts

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Abstract

Cloud storage has become an essential platform for storing and sharing large volumes of data in industrial environments. However, ensuring secure data sharing while preserving data privacy and usability remains a significant challenge. Traditional encryption techniques provide strong security but limit efficient data retrieval, especially when dealing with encrypted data. This project focuses on developing a secure data sharing framework using attribute-based encryption (ABE) combined with searchable encryption techniques to enable efficient and privacy-preserving data access in cloud environments. The proposed system introduces a dual-server architecture based on Dual-Server Public Key Authenticated Encryption with Keyword Search (DPAEKS) to enhance security against internal keyword guessing attacks and unauthorized access. The system allows data owners to upload encrypted files with associated keywords, while data users can securely search and retrieve relevant data without revealing sensitive information. Attribute-based access control ensures that only authorized users with matching attributes can decrypt and access the data. The use of two non-colluding servers improve privacy protection and reduces the risk of key leakage and attack vulnerabilities. Experimental evaluation demonstrates that the proposed system improves search efficiency, accuracy, and overall security compared to existing approaches. The framework provides a practical and scalable solution for secure cloud data sharing in industrial contexts, ensuring confidentiality, integrity, and efficient data utilization.

Keywords: *Cloud Storage, Attribute-Based Encryption (ABE), Searchable Encryption, Dual-Server Architecture, Secure Data Sharing*

MSEC-RS-2026-IT-011

Self-Regulated AI Agent with Goal Conflict Resolution

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Abstract

Addiction rehabilitation requires balancing shifting psychological states and competing motivations while requiring the person to exert self-control over a sustained period of time. Conventional AI systems just do not have those specialties. This work is being proposed as a framework for a self-regulated AI agent that comprises the dynamics of a multi-agent system. This multi-agent system has four specialized agents: Crave agent (cravings agent), Therapeutic agent (therapeutic engagement), Well-being agent (well-being), and social stability agent. These agents share the environment and, in being contextually aware of activities taking place in each other's environment, continue to be proposed in their actions. The actions of these agents co-exist with a Meta-Regulator that resolves agent conflicts to respond to method or behavioral action selection. It uses aforementioned principles such as goals inhibition; delay of gratification and value prioritization to resolve conflicts. Thus, the agents of the system incorporate habit behavioral changes while bringing in Multi-Agent Reinforcement Learning techniques tuning data in as conditions of behavioral disposition and signals from the environment shift. Simulation experiments found that the model can deliver balanced yet personally recommended action responses to mimic recovery decision-making while also considering and perceiving transparency in decision-making. This work indicates the potential of such architectures, patterned after MARL principles embedded in psychological science, in behavioral recovery practice that respects the requirement for adaptive, ethical behavior.

Keywords: *Self-regulation, Multi-Agent Reinforcement Learning, Goal conflict resolution, Addiction rehabilitation, MetaRegulator, Adaptive decision-making*

MSEC-RS-2026-ME-001

UV-A Induced Photocatalytic Degradation of Acid Green 16 Using La2 WO6 /ZnO Heterojunction Nanocomposites: Synthesis and Biological Applications

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Abstract

Increasing concerns related to dye-contaminated wastewater and the demand for multifunctional materials have prompted studies on semiconductor-based nanocomposites for environmental and biological applications. In this context, a 7 wt.% La2 WO6 /ZnO nanocomposite was synthesized via a hydrothermal method and examined for its structural, optical, photocatalytic, and biological characteristics. SEM analysis showed the coexistence of smaller ZnO particles and comparatively larger La2 WO6 particles within the same microstructure. XRD confirmed crystalline hexagonal ZnO and monoclinic La2 WO6 phases without detectable structural alteration. UV–DRS measurements showed band gap values of 3.2 eV for ZnO, 2.8 eV for La2 WO6 , and 2.6 eV for the composite. Photocatalytic performance was evaluated through the degradation of the anionic azo dye Acid Green 16 (AG-16) under UV light irradiation. The influence of solution pH, initial dye concentration, and catalyst dosage on degradation efficiency was systematically investigated, and the catalyst was reused for five successive cycles under identical experimental conditions to assess its stability. Antibacterial activity was assessed using the agar well diffusion method, and cytotoxicity was evaluated against MCF-7 breast cancer cells. The results demonstrate the structural features and functional performance of the La2 WO6 /ZnO nanocomposite in AG-16 degradation, along with its biological activity under the studied conditions.

Keywords: *antibacterial activity, anticancer activity, anti-inflammatory activity, La2 WO6, ZnO nanocomposites, photocatalysis, wastewater treatment*

MSEC-RS-2026-ME-002

Optimization of Aircraft Main Landing Gear Shock Strut

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Abstract

The Landing gear is one of the critical subsystems of an aircraft. In case of failure in the landing gear or any of its subcomponents, there will be disastrous consequences. The Landing gear consists of plenty of sub-systems in it. But out of all one of the most important subsystems if not the only important subsystem is the shock strut. It does the same work as the spring and damper in cars, it stops the vibrations and absorbs the landing forces, and dissipates before it is transmitted to the chassis. Here instead of spring, pre-charged gas is used to get the spring effect, and hydraulic oil is used to achieve the damping effect. And it is called Oleo pneumatic shock strut. The word oleo has its etymology in Latin where oleum - means oil and the word pneumatic means something related to air. So, oleo pneumatic essentially means oil-air shock strut. These shock struts are the heart of the landing system but they also have high mass. Since the Landing Gear of an aircraft accounts for 5-6% of the weight of the aircraft, it is necessary to reduce the weight of the Landing Gear. The project aims to design the shock strut, analyze different materials which can be used in an MLG shock strut instead of conventional materials, and select the most suitable material which can provide similar performance to conventional materials at a comparatively lesser weight. It also includes enhancing the dissipation of shock and impact loads on the MLG oleo strut and optimizing orifice design.

It is carried out by modeling an aircraft shock strut using 3D CAD software and comparing the FE Analysis values of various alternative materials using an Analysis Software such as Ansys. This Analysis can be used to find out which material has better Stress, strain, and deformation-withstanding capabilities and which material offers better weight saving and weight reduction.

Keywords: *Aircraft Shock Strut, Landing Gear, 3D Modelling, Material Selection, Ansys.*

MSEC-RS-2026-ME-003

CFD Analysis of Aerodynamic Brake Rotor Cooling Enhancement Using Transient Thermal Simulation

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Abstract

Brake rotors in passenger vehicles experience significant thermal loading during braking due to friction between the brake pad and disc surface. More temperature rise may lead to thermal accumulation and reduce the life of braking components. Effective cooling of the brake rotor is essential to maintain safe operating conditions. This study investigates the effect of aerodynamic modification of a passenger vehicle front body on airflow distribution near the brake rotor using Computational Fluid Dynamics (CFD). A sedan vehicle model was analysed using ANSYS Fluent to evaluate airflow behaviour around the wheel region. The baseline configuration was compared with a modified geometry designed to guide airflow toward the brake rotor. The result indicates that the airflow velocity near the brake rotor increased from approximately 5 m/s to about 12 m/s after modification. Transient thermal analysis of the brake disc was performed using ANSYS Mechanical with a time dependent heat flux representing braking conditions. The modified configuration reduced the peak rotor temperature from 72.03°C to 68.76°C, corresponding to a temperature reduction of approximately 4.1%. The study demonstrates that aerodynamic optimization of the vehicle body can enhance brake rotor cooling without negatively affecting overall vehicle aerodynamics.

Keyword: *Brake Rotor Cooling, Computational Fluid Dynamics (CFD), Aerodynamic Optimization, Passenger Vehicle, ANSYS Fluent, ANSYS Mechanical.*

MSEC-RS-2026-ME-004

Influence of Enhanced Wind Speed on Heat Transfer in Alumina/De-Ionized Water Nanofluid Based Parabolic Trough Collectors: An Experimental and Analytical Study

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Abstract

Among the various solar energy harnessing techniques, the parabolic trough solar collector (PTSC) remains one of the most established and widely implemented technologies for both domestic and industrial applications. Ongoing research efforts have largely been directed toward improving the performance of PTSC systems through design modifications and optimization strategies. In contrast, the present study provides an analytical investigation of heat transfer performance under enhanced wind speed conditions, utilizing experimental data for validation. Specifically, the impact of a 5% increase in wind speed on the thermal behaviour of the collector is examined for all alumina/deionised water nanofluid from 0% to 4.0% @ 0.5% and all mass flow rate from 0.02kg/s to 0.06 kg/s with a increment of @ 0.01 kg/s. Reynolds number and Nusselts number were increased in an average of 5.01% and 2.967%. Also finally the loss coefficients of wind, total heat and overall heat were increased averagely in the range of 2.973%, 0.376% and 0.377% respectively. Unlike previous studies that primarily emphasize structural or material alterations, this work highlights the influence of external convective effects, offering new insights into performance variation attributable to environmental factors.

Keywords: *PTSC, alumina, di water, wind velocity enhancement, heat transfer*

MSEC-RS-2026-ME-005

Numerical Investigation of Energy Consumption and Stress Evolution During Drilling of Eggshell-Reinforced Epoxy Composites Using DEFORM-3D

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Abstract

The predictive simulation platform Deform 3D compare machining performance, material removal rates, and surface coarseness. Trials employ Epoxy resin-5% Egg Shell and Epoxy resin-10% eggshell polymer composites. G power calculations in SPSS ensure 80% pre-test power by estimating sample sizes. Both groups of each composite specimen undergo vertical drilling at 500, 700, 900, and 1100 m/min, 0.08, 0.12, 0.16, and 0.2 mm depths, and 0.5, 0.9, 1.3, and 1.7 mm/rev feed rates. The L16 orthogonal array of Taguchi's approach optimum cutting conditions showed that rotating speeds and feed rates dominate surface roughness and material removal rates. The optimal surface roughness (SR) of 0.600 µm is achieved at 900 m/min and 0.5 mm/rev feed rate, while the greatest material removal rate (MRR) is achieved at 700 m/min, obtaining 0.280 mm^3/min. MRR and surface roughness amplified by 9% and 27% by the values of 0.27 mm3/min and 0.316 µm respectively. Statistical analysis using SPSS indicated substantial improvements ($p< 0.05$). The research found that eggshell powder increases epoxy composite MRR and surface roughness by 30 35%. Furthermore, deform 3D simulation findings show a complicated relationship between drilling speed and material behaviour. Higher speeds (900 and 1100 m/min) use less energy but raise stress and strain, whereas intermediate speeds like 700 m/min use more. These outcomes are used to tune drilling operations to achieve optimal energy efficiency, tool wear, and residual material integrity; this improves both the performance and life of the drilling tool and lightweight polymer composite workpiece.

Keyword: *Epoxy composites reinforced with eggshell particles, material removal rate analysis, Deform 3D simulation, Taguchi method, and statistical evaluation.*

MSEC-RS-2026-ME-006

Experimental Investigation on Impact Behavior of Copper Wire, Al_2O_3, and SiC Particle Embedded Glass Fiber Reinforced Composites

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Abstract

The incorporation of copper wire and ceramic reinforcements such as Al_2O_3 and SiC significantly influences the internal structure and load transfer capability of the GFRP composites. The variation in pitch distance plays a crucial role in determining the distribution of stresses and energy absorption characteristics under impact loading. A smaller pitch distance enhances the reinforcement density, leading to improved bonding and higher resistance to crack propagation, whereas larger pitch distances may reduce stiffness but contribute to better flexibility. Additionally, the presence of Al_2O_3 and SiC particles improves surface hardness and wear resistance, thereby enhancing durability. The comparative analysis between bi-directional and unidirectional laminates further highlights the effect of fiber orientation on mechanical behavior, where bi-directional laminates generally exhibit better multidirectional strength and impact resistance, while unidirectional laminates provide superior strength along the fiber direction.

Keyword: *Glass Fiber reinforced Composites, Copper wire, Al2O3, SiC*

MSEC-RS-2026-ME-007

Design, Development and Implementation of a Kanban Based Pattern Production System for Spline Tube Production Line

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Abstract

In multiproduct manufacturing environments, production planning and material flow management become complex due to frequent product changeovers and varying customer demand. High-mix, low-volume production systems often face challenges such as excess work-in-progress (WIP), poor space utilization, and material congestion on the shop floor. Traditional push-based production systems rely on demand forecasts, often resulting in overproduction, increased inventory costs, and reduced responsiveness to customer requirements. Lean manufacturing principles help address these issues by focusing on waste reduction and efficient production flow. The Kanban system is an effective lean tool used to implement pull-based production control. This project focuses on the design and implementation of a Kanban- based pattern production system in a spline tube manufacturing line. The production process includes Four Lobe Formation, CNC machining, and VMC operations, handling two major product families: 20 PCD and 25 PCD spline shafts with multiple variants. A detailed shop floor study was conducted to analyze the existing layout, workflow, material handling practices, and WIP levels. Based on this analysis, a structured Kanban system was designed using parameters such as demand rate, lead time, container capacity, and safety factor. A pattern production sequence was introduced to ensure balanced production of multiple variants. Kanban cards were laminated for durability, and Kanban posts were installed at key locations such as WIP areas and inventory zones. The implementation improved material visibility, optimized space utilization, and resulted in smoother production flow and improved operational efficiency.

Keyword: *Kanban system, lean manufacturing, pattern production, pull-based production, push-based system, work-in-progress reduction.*

MSEC-RS-2026-ME-008

Design and Impact Analysis of Hybrid Foam-Filled Voronoi Sandwich Panels for Vehicle Crumple Zones

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Abstract

In order for a car to be crash-worthy, its components must be light in weight and efficient at absorbing the impact of a crash. In a conventional sandwich panel with a uniform honeycomb core, high peak impact forces and irregular collapse modes are often observed under dynamic axial loading, thus limiting their use in car crumple zones. To overcome these limitations, this paper proposes the design and impact response analysis of a hybrid foam-filled Voronoi sandwich panel, which is modeled after natural cellular structures. Nine different Voronoi sheet layouts were systematically developed using the Taguchi design approach & Rhino Software and analyzed using static structural analysis to determine an optimal cellular design. With the optimized design, suitable materials for the Voronoi sheets and foam core were chosen, and a hybrid multi-layer sandwich panel was designed. Explicit dynamic finite element analysis was carried out to analyze the impact response of the proposed panel under axial loading conditions simulating car crumple zones. The analysis results show that the foamfilled Voronoi sandwich panel has superior progressive collapse characteristics, lower peak impact force, and better energy absorption capacity than the conventional uniform-core sandwich panel. These results demonstrate the feasibility of hybrid Voronoi-foam sandwich panels as efficient energy-absorbing structures in car passive safety systems.

Keyword: *Mechanical engineering, Automotive, bio-inspired structures, crumple zone, crashworthiness, modeling and simulation.*

MSEC-RS-2026-ME-009

Combined effect of Carbon nanotube and toroidal re-entrant chamber with modified fuel injection pressures: A study in agricultural DI diesel engine

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Abstract

The present experimental study investigates the effect of injection pressure and toroidal re-entrant combustion chamber geometry in a single-cylinder diesel engine fuelled with oxygenated fuel blends enhanced with carbon nanotubes (CNT). Since oxygenated fuels generally possess inferior physico-chemical properties compared to conventional diesel, 20 ppm of CNT additives are incorporated to improve ignition characteristics and promote catalytic oxidation. The resulting mixture is termed as Oxygenated Fuel Blend (OFB). Experiments were conducted by operating the high-performance fuel at injection pressures of 18, 20, 22, and 24 MPa, while maintaining a constant injection timing of 22° bTDC in the toroidal re-entrant chamber. The results indicate that an injection pressure of 22 MPa provides optimal performance, achieving a maximum brake thermal efficiency (BTE) of 35.5% and a minimum brake specific energy consumption (BSEC) of 10.13 MJ/kWh. These improvements are mainly attributed to enhanced atomization, better evaporation, intensified swirl motion, and improved air–fuel mixing. Emission analysis revealed reductions in hydrocarbon (HC), carbon monoxide (CO), and smoke emissions, with slight increases in NOx and CO_2. Combustion parameters such as cylinder pressure, heat release rate (HRR), and cumulative heat release rate (CHRR) also improved, indicating enhanced combustion efficiency.

Keyword: *Re-entrant; CNT; injection pressure; performance; combustion; emission*

MSEC-RS-2026-ME-010

Analysis and Optimisation of Nanofluid Based on Parabolic Trough Solar Collector

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Abstract

Solar-to-thermal energy conversion is an important approach for utilizing renewable the present work focuses on improving solar-to-thermal energy conversion using nanofluids in a Concentrated Solar Power (CSP) system. A parabolic trough collector is considered, as it is widely used for efficient solar energy utilization. Conventional absorber coatings suffer from reduced efficiency at higher temperatures due to increased heat losses. In this project, solar absorbing nanofluids are used as an alternative to enhance heat transfer and energy absorption. The nanofluid, consisting of base fluid mixed with nanoparticles, enables volumetric absorption of solar radiation, improving overall system performance. Key parameters such as nanoparticle concentration, fluid temperature, and solar intensity are analyzed to study their effect on efficiency. The results show that optimized nanofluid properties significantly improve thermal performance and reduce energy losses compared to conventional methods. Hence, the use of nanofluids in parabolic trough systems is identified as an effective solution for enhancing solar energy conversion and supporting sustainable energy applications.

Keyword: *Mechanical engineering, Automotive, bio-inspired structures, crumple zone, crashworthiness, modelling and simulation.*

MSEC-RS-2026-ME-011

Influence of Injection timing on an Oxygenated Biodiesel Blend (OBB) Operated in an Agricultural Stationary Diesel Engine

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Abstract

The present work focuses on analysing the influence of Injection timing (advanced and retarded) when fuelled with oxygenated biodiesel blend (also known as OBB). The oxygenated biodiesel blend is prepared by blending 20% Biodiesel, 70% diesel and 10% ethanol fuel. The OBB blend is a ternary fuel, i.e a mixture of three fuels. This proposed fuel aims to lower the dependency of the fossil fuels and lower the fuel prices. The oxygenated biodiesel blend (OBB) were subjected to various injection timings from 21° to 24° with increments of 1°, among which 23°bTDC was considered as standard injection timing, 22°bTDC and 21°bTDC were retarded injection timings. 24°bTDC was considered as advanced timing. Performance characteristics like BTE and BSEC, emission characteristics like HC (Hydrocarbon), CO (Carbon monoxide), NOx (Nitrogen oxides) and smoke along with combustion characteristics like cylinder pressure, HRR (Heat release rate) and CHRR (Cumulative heat release rate) were considered for analysis. Based on experimentation, it is observed that, OBB-22 ° IT resulted in highest BTE of 33.8% and lowest BSEC of 10.59MJ/kW-hr indicating that OBB operates effectively on retarded injection timing. OBB-22° IT has lowered HC and CO emissions by 9.18% and 16.83% with only at a compensation of higher NOx in comparison with standard timing of 23°bTDC. 22 ° IT stays at acceptable range of cylinder pressure and HRR at 75.42bar and 85.34J/ ° CA in comparison with other injection timings. Hence, owing to enhanced performance, combustion and emissions, OBB-22 ° IT is considered as effective injection timing blend for improved performance and lowered emissions.

Keyword: *HRR (Heat release rate), CHRR (Cumulative heat release rate), oxygenated biodiesel blend (OBB)*

MSEC-RS-2026-ME-012

Analysis and Simulation of Solar Powered Battery Hybrid Vehicle

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Abstract

A solar battery hybrid vehicle is an eco-friendly transportation system that integrates solar energy generation with battery storage to power an electric motor, thereby reducing dependence on fossil fuels. This study presents the design and analysis of a solar-assisted hybrid vehicle system that utilizes photovoltaic panels to convert sunlight into electrical energy through the photovoltaic effect. The generated energy is stored in a 3.7 V rechargeable battery, which ensures continuous operation during low or no sunlight conditions. A buck converter is employed to regulate and step down the voltage to a suitable level for the DC gear motor, enhancing system efficiency and protecting electrical components. The motor converts electrical energy into mechanical motion to drive the vehicle. The system also incorporates basic control circuits for efficient power management. This integrated approach combines energy generation, storage, and utilization, demonstrating key principles of renewable energy engineering. The proposed hybrid system enhances reliability, improves energy efficiency, and promotes sustainable transportation by reducing emissions and fuel consumption.

Keyword

Solar Hybrid Vehicle, Photovoltaic Effect, Renewable Energy, GFRP, Buck Converter, DC Gear Motor, Energy Storage.

MSEC-RS-2026-TAMIL-001

அறிவியல் மற்றும் ததொழில் நுட்பத்தில் நிலலயொன வளர்ச்சி விவசொய அறிவியல்

முனைவர் சு.சித்ரா, உதவிப் பபராசிரியர். தமிழ்த்Fனை.
மீைைாட்சி சுந்தரராஜை் பபாறியியல் கல்லூரி,
பகாடம்பாக்கம், பெைை்னை - 24.

ஆய்வுச் சுருக்கம் :

விவொய அறிவியல் Fனையில் ஆய்வு பமற்பகாள்ளுவதைக்கு பல முக்கிய காரணங்கள் உள்ளை. குறிப்பாக உணவுப் பாFகாப்பு, ைுபத்தி பெயல்திைை் மைற்றும் சுைை்றுெை்சூழல் நினலத்தைனம ஆகியவைனை உறுதி பெய்ய இந்த ஆய்வு அவசியமாகிைF. இனவ மண் ஆபராக்கியம் பயிர் சுழைசி, Fல்லிய விவொயம் பபாைை்ை புFனமகனளப் பரப்பி ைுபத்தித்திைனை அதிகரிக்கிைF. பூெை்சி கட்டுப்பாடு நீர் பமலாண்னம ஆகியவைற்றில் பமம்பாடுகனள ஊக்குவிக்கிைை்ைை. காலநினல மாைை்ைம், பல்லுயிர் பாFகாப்பு பபாைை்ை ெவால்களுக்கு தீர்வுகனள வழங்கி, நினலயாை விவொயத்னத ஆர்வமூட்டுகிைை்ைை. உயர் வினளெ்ெல் தரும் வினத வனககள், பெயைனக உரங்கள், பூெை்சிக்பகால்லிகள் பபாைை்ைவைற்றிைை் பயை் பாட்னட விளக்குதல், வாைினல முைை்ைறிவுப்பு, மண் பொதனை பபாைை்ை சுைை்றுெை்சூழல் அறிவியல் ஆகியவைனை எளினமயாக விளக்குதல் அவசியமாகிைF. இF பபாைை்ை ஆய்வுகள் பாரம்பரிய விவொயத்னத அறிவியல் முனைகளுடை் இனணத்F உணவுப் பாFகாப்னப உறுதி பெய்கிைை்ைை. நாட்டிைை் பபாருளாதார வளர்ெை்சிக்கு விவொயத்திைை் முFபகலும்பு பங்னக வலுப்படுத்Fகிைை்ைை. விவொய அறிவியலில் அறிவியல் மைற்றும் பதாழில்நுட்பங்கள் நினலயாை வளர்ெை்சினய ஊக்குவிக்கும் முக்கிய கருவிகளாக உள்ளை இனவ ைுபத்தினய அதிகரித்F, வளங்கனள சிக்கைமாக பயைபடுத்தி, சுைை்றுெை்சூழனலப் பாFகாக்க உதவுகிைை்ைை. Fல்லிய விவொயம் பெயைனக நுண்ணறிவு (AI) மைற்றும் ட்பராைை்கனளப் பயைபடுத்தி மண், வாைினல மைற்றும் பயிர் ஆபராக்கியத்னத கண்காணிக்கிைF. இF நீர், உரம் மைற்றும் பூெை்சிக்பகால்லிகனள ெரியாை அளவில் பயைபடுத்தி பெலவுகனளக் குனைக்கிைF. அFமட்டுமிைை்றி வினளெ்ெனல அதிகரிக்கிைF.

முக்கியச் தசொற்கள்: பெயைனக நுண்ணறிவு, விவொய அறிவியல் ட்பராைை், பெயைனக உரங்கள்.

MSEC-RS-2026-MATHEMATICS-001

An Integrated Interval-Valued Intuitionistic Fuzzy Transportation–Pert Model for Cost–Time Optimization Under Uncertainty

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Abstract

Real-world transportation and logistics systems are always affected by uncertainty, imprecision, and insufficient information. This is because of things like changing demand, congestion, and changes in the environment. This paper provides an integrated framework that combines the Interval-Valued Intuitionistic Fuzzy Transportation Problem (IVIFTP) with the Program Evaluation and Review Technique (PERT) to fully optimise costs and time.

At first, interval-valued intuitionistic fuzzy numbers are used to estimate transportation costs in order to accurately reflect uncertainty and reluctance. A strong ranking-based defuzzification method is used to turn fuzzy parameters into clear ones. Then, a Modified Allocation Table Method (ATM) is used to find the best transportation plan. After that, the best allocations are put into a PERT network, where each transport task is described using time estimates that are based on chance. This makes it possible to figure out the expected lengths of activities, the variations, and the critical path.The suggested hybrid model not only cuts down on transportation costs, but it also gives useful information on project scheduling, schedule unpredictability, and risk assessment.The results show how well combining fuzzy optimisation with probabilistic scheduling works. This is a powerful tool for making decisions in smart logistics, supply chain management, and real-time transportation systems that work in uncertain situations.

Keywords: *IVIFTP, PERT, Fuzzy Optimization, Transportation Problem, Uncertainty Modeling, Smart Logistics*

MSEC-RS-2026-MATHEMATICS-002

A Computational Framework for Solving Multi-Variable Diophantine Equations Using Modular Arithmetic and Euclidean Algorithms

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Abstract

Solving multi-variable Diophantine equations is a significant challenge in discrete mathematics, with applications in cryptography, combinatorial optimization, and integer programming. As the number of variables and polynomial degree increase, classical methods and exhaustive searches become impractical due to exponential growth in the solution space. This highlights the need for scalable approaches that maintain mathematical rigor.This work introduces a hybrid computational framework that combines modular arithmetic, the Extended Euclidean Algorithm, and constraint reduction techniques into a structured solution process. It begins with modular filtering to eliminate infeasible congruence classes, reducing the search domain before computation. Divisibility conditions are enforced using greatest common divisor analysis, while complex equations are simplified into equivalent lower-complexity forms, ensuring a finite and manageable solution space.Theoretical analysis shows that the framework removes up to 80% of infeasible candidates before enumeration, reducing computational complexity to O(n log n) for linear systems compared to higher costs in brute-force methods. The framework was validated on linear, bilinear, and quadratic Diophantine problems, consistently producing correct integer solutions with analytical clarity.Unlike heuristic approaches, this method guarantees correctness and provides a reproducible process applicable to cryptography, discrete optimization, and integer system modeling, offering a scalable foundation for solving complex integer equations efficiently.

Keywords *Diophantine equations, integer solutions, modular arithmetic, Extended Euclidean Algorithm, constraint reduction, computational complexity*

MSEC-RS-2026-MATHEMATICS-003

Non-Parametric Density Estimation for Predictive Congestion Control in High-Velocity Edge Networks

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Abstract

Managing resources in edge computing is challenging due to the bursty nature of Internet of Things (IoT) data, which often conflicts with traditional statistical models. Existing congestion control methods rely on parametric statistics, such as Poisson and Gaussian distributions, but fail to capture heavy-tailed behavior and sudden traffic spikes. This leads to issues like buffer bloat and packet loss, especially in resource-constrained environments. This paper proposes a non-parametric predictive framework based on Kernel Density Estimation (KDE) to model packet arrivals more accurately. By avoiding fixed distribution assumptions, the system constructs a real-time probability density function for incoming traffic. A key component is an automated bandwidth selection mechanism that balances sensitivity and computational efficiency. The method uses a sliding window to collect recent packet arrival timestamps, which are processed using a Gaussian kernel to produce a smooth probability distribution. This approach overcomes discretization issues and enables early detection of changes that may indicate congestion. To ensure efficiency, a plug-in estimator determines bandwidth while minimizing computation. By bridging statistics and network engineering, the framework supports adaptive traffic management, enabling low-power devices to handle traffic surges effectively and improving system resilience.

Keywords: *Non-parametric statistics, Kernel Density Estimation (KDE), bandwidth optimization (h), PDF modeling, Poisson distribution, M/M/1 queuing, edge computing, buffer bloat, stochastic traffic.*

MSEC-RS-2026-MATHEMATICS-004

Adaptive Chaos-Driven Homomorphic Encryption

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Abstract

This paper presents an Adaptive Chaos-Driven Homomorphic Encryption (ACDHE) framework that integrates nonlinear dynamical systems with modern cryptographic techniques to enhance secure data processing. The model uses chaotic maps with sensitivity to initial conditions, ergodicity, and deterministic randomness to generate pseudo-random cryptographic keys. Unlike traditional methods, the chaotic control parameter is dynamically adapted based on input data and ciphertext noise, enabling flexible and data-driven key generation. The encryption scheme is based on ring-based homomorphic encryption, using polynomial rings and finite field operations to support computation on encrypted data without compromising privacy. By combining chaotic mappings with algebraic structures such as modular arithmetic, lattice constructs, and ring homomorphisms, the framework ensures strong mathematical rigor and security. Experimental and analytical results demonstrate improved entropy, reduced statistical predictability, and enhanced resistance to cryptographic attacks while preserving correctness of homomorphic operations. The framework also supports adaptive traffic management, enabling localized servers and low-power devices to efficiently handle data surges, improving resilience, scalability, and overall system performance.

Keywords: *Adaptive Chaos, Homomorphic Encryption, Polynomial Rings, Finite Fields, Ring Homomorphism, Lattice-Based Cryptography, Nonlinear Dynamical Systems, Modular Arithmetic, Secure Computation, Cryptographic Algebra*

MSEC-RS-2026-MATHEMATICS-005

Performance Analysis of Information Set Monte Carlo Tree Search in Phantom Tic-Tac-Toe

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Abstract

This study investigates the application of Information Set Monte Carlo Tree Search (IS-MCTS) to Phantom Tic-Tac-Toe, a partially observable variant of the classic zero-sum game. Unlike standard Tic-Tac-Toe, the Phantom version introduces "fog of war," hiding opponent moves and transforming the game into a Partially Observable Markov Decision Process (POMDP). To address this uncertainty, an IS-MCTS agent using a determinization-based approach was developed. The agent generates multiple plausible game states consistent with the current information set and evaluates move outcomes in terms of win, loss, or draw probabilities. Move selection is guided by the UCB1 policy, balancing exploration and exploitation over 1,000 iterations per decision to identify optimal actions. Performance was evaluated through a 100-game tournament against a random baseline agent. The IS-MCTS agent achieved a 65%win rate and 7% draw rate, significantly outperforming the baseline's 15%win rate. The agent effectively infers hidden opponent positions using move collisions and turn history. Additionally, the method demonstrates consistent computational efficiency, enabling real-time execution on standard hardware. Overall, IS-MCTS proves effective in handling hidden information through probabilistic reasoning and scalable decision-making under uncertainty.

Keywords: *Information Set Monte Carlo Tree Search(IS- MCTS), Determinization, Partially Observable Markov Deci- sion Process(POMDP), Imperfect Information Games,UCB1 Algorithm ,Belief-State Management, Phantom Tic-Tac-Toe.*

MSEC-RS-2026-MATHEMATICS-006

Graph-Based Adaptive Clustering in Wireless Sensor Networks Using Convex Optimization for Network Lifetime Maximization

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Abstract

Wireless Sensor Networks (WSNs) are resource-constrained systems where energy efficiency directly determines network lifetime. This paper proposes an adaptive graph-based clustering framework integrated with convex optimization to maximize network operational duration by minimizing total energy consumption while maintaining network connectivity. Unlike fixed clustering protocols, our adaptive mechanism dynamically reconfigures network topology based on residual energy and node connectivity patterns, automatically selecting optimal cluster heads and cluster memberships across communication rounds. Extensive simulations demonstrate that the proposed method extends network lifetime by up to 35% compared to traditional approaches like LEACH, while achieving more balanced energy utilization across all nodes. The framework is particularly effective for IoT applications, environmental monitoring, and large-scale distributed sensing deployments.

Keywords: *Wireless Sensor Networks, Energy Efficiency, Adaptive Clustering, Graph theory, Convex Optimization, Network Lifetime, IoT, Load Balancing*

MSEC-RS-2026-MATHEMATICS-007

Soret Effects on Radiative Heat and Mass Transfer Flow of a Rotating Fluid in The Presence of Magnetic Filed

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Abstract

This article examines the effects of Hall current, radiation, and Soret phenomena on heat and mass transfer in magnetohydrodynamic flow over an infinite vertical plate in a rotating fluid. The system considers uniform rotation about the plate's normal axis and a transverse magnetic field applied into the fluid region. With a low magnetic Reynolds number, the induced magnetic field is neglected. The governing equations of momentum, energy, and mass diffusion are transformed into non-dimensional form using similarity transformations. These equations are solved analytically using the Laplace transform technique and numerically using MATLAB. Expressions for primary and secondary velocities, temperature, and concentration profiles are derived. A deep-learning model is further trained on generated data to validate and predict flow behavior. Results show that increasing the Prandtl number reduces thermal diffusion, while higher Soret numbers enhance mass transfer. The influence of key parameters such as Hall parameter, Hartmann number, rotation parameter, Schmidt number, Prandtl number, Soret number, and thermal and mass Grashof numbers is analyzed. Graphical results illustrate variations in velocity, temperature, and concentration. The study has applications in geosciences, chemical engineering, and other fields involving heat and mass transfer processes.

Keywords: *Soret Effect, Magnetic Field, Hall parameter, Hartmann number, Schmidt number, Prandtl number, Thermal Grashof number, Mass Grashof number*

MSEC-RS-2026-PHYSICS-001

Experimental Study on the Influence of pH and Temperature in a Mini Mesophilic Anaerobic Digester for Sustainable Energy Production

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Abstract

Biogas is a renewable source of energy, playing a vital role in the development of sustainable energy production, waste management, reduction of environmental pollution, and efficient use of resources. In this study, a mini mesophilic anaerobic digester was designed and analyzed in order to determine the influence of pH values and temperature on the production of biogas. Organic food waste was used as the substrate in the experiment, collected from households as waste. The experiment was carried out using a 100-L biogas digester, operating in the mesophilic range (30-40°C) for a period of 10 days. The pH values of the substrate were measured on a daily basis, as were the temperatures, while the evaluation of the produced biogas was carried out using a water boiling test, with the heat energy released during combustion calculated using the concept of the change in enthalpy. The findings show that the maximum amount of biogas was produced when the pH values were near neutral, i.e., around 7, with the temperature ranging from 33-34°C. The study shows that the pH values and temperatures have a significant influence on the efficiency of the production of biogas.

Keywords: *Food waste*, *Biogas, Mesophilic digestion, methane*

MSEC-RS-2026-PHYSICS-002

Comparison of Ground Level Ozone Concentration During Winter and Summer at an Industrial Area, Chennai

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ABSTRACT
Tropospheric ozone is also known as Surface Ozone (SOZ) or Ground Level Ozone (GLO). O_3 is also a major component of photochemical smog and is a well-known hazardous element for human health. Ground level ozone is not directly emitted into the atmosphere. The Ozone concentration is influenced by the intensity of solar radiation and chemical reaction between oxides of nitrogen (NOx) and volatile organic compounds (VOC) in the presence of sunlight. This paper attempts to study the variation of ground level ozone for two seasons summer (Mar.' – May.'16) and winter (Jan.'-Feb.'16) at Industrial area, Chennai. The overall result shows that lowest concentration of ground level ozone was observed in winter as compared to the summer season.

Keywords: *Ground level ozone, A series 500, industrial Area, metrological parameters.*

MSEC-RS-2026-PHYSICS-003

Integration of Artificial Intelligence in Nanotechnology: From Prediction to Application

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ABSTRACT

The review provides an extensive overview of artificial intelligence-based integration in nanoparticle research and development. Nanoparticles have specific physicochemical properties resulting from their nanoscale dimensions, which enable a variety of applications in medicine, catalysis, environmental remediation, sensing technologies, and energy storage. Nevertheless, nanoparticle synthesis conditions are complex to optimize owing to a large number of experimental variables involved. Artificial intelligence-based approaches such as machine learning, deep learning, and reinforcement learning offer efficient data-based approaches to predict nanoparticle properties. Moreover, artificial intelligence-based approaches can be employed to optimize nanoparticle synthesis conditions. This review provides an overview of recent artificial intelligence-based nanoparticle discovery approaches. It covers major nanoparticle types, methodologies, and applications. Moreover, artificial intelligence-based nanoparticle discovery approaches are discussed in terms of limitations. Finally, future research directions are discussed.

Keywords: *Nanoparticles, Artificial Intelligence, Machine Learning, Nanotechnology, Materials Informatics, Nanomedicine.*

MSEC-RS-2026-PHYSICS-004

Synthesis of Rice Husk Activated Carbon and MnO_2-Infused Porous Activated Carbon Variants for Capacitive Desalination

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Abstract

Agriculture waste is a renewable and abundant resource that is naturally carbon-rich. In this report, carbon is prepared from rice husks and treated with Sulphuric acid (H_2SO_4). Rice husk activated carbon (RHAC) electrodes with a porous nature were synthesized by a simple and effective chemical activation method. XRD patterns of rice husk-activated carbon displayed standard graphitic peaks, while those of MnO_2 decorated rice husk activated carbon (MnO_2/RHAC) revealed crystalline nature. The SEM analysis indicates that the average pores are 6 nm in size, which is consistent with the BET results. The MnO_2/RHAC electrode that was fabricated exhibited an outstanding specific capacitance of 385 Fg^{-1}, surpassing most of the values reported in prior literature. In 600 mL of NaCl solution at 1.2V, the salt adsorption rate and capacity are determined to be 0.85 mg/g/min and 21 mg/g, respectively. This report shows that rice husk-activated carbon can be used in the treatment of brackish water desalination to remove salt and heavy metal ions.

Keywords: *Activated Carbon, Manganese dioxide, Metal ion removal, Rice husk, Water desalination*.

MSEC-RS-2026-PHYSICS-005

Quantum-Thermal Optimization in Sub-10nm Architectures: Overcoming the 'Heat Wall' via Ballistic Transport in Graphene-Based Cooling Systems

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Abstract

In the era of sub-10 nm semiconductor fabrication, smartphone processors have encountered a critical thermal bottleneck known as the "Heat Wall," where classical cooling methods fail to dissipate localized heat flux effectively. This research investigates the failure of traditional metallic heat sinks through the lens of Classical Free Electron Theory (Drude Model), which characterizes charge transport as a series of stochastic collisions between electrons and the atomic lattice. These collisions, driven by the high Effective Mass of charge carriers in standard conductors, generate significant kinetic friction and entropy, leading to parasitic thermal dissipation, performance throttling, and accelerated chemical degradation of lithium-ion batteries. To transcend these classical limitations, this paper proposes a paradigm shift toward Quantum-Thermal Management using Graphene. Utilizing Quantum Field Theory and the Sommerfeld Model, we analyze the behavior of electrons within Graphene's hexagonal honeycomb lattice, where they behave as massless Dirac fermions. We demonstrate that Graphene facilitates "ballistic transport," allowing charge carriers to bypass the scattering mechanisms prevalent in traditional metals. By maintaining a constant Fermi velocity ($\sim 10^6$ m/s) with near-zero resistance, Graphene-based architectures significantly minimize phononic heat generation.Our findings suggest that integrating graphene thin-films into next-generation smartphone SoCs (System on Chips) not only maintains peak clock speeds under heavy computational loads but also extends battery longevity by maintaining an isothermal internal environment. This research provides a theoretical and practical framework for moving beyond classical material constraints into the quantum era of mobile hardware design.

Keywords: *Graphene, Dirac Fermions, Ballistic Transport, Drude Model, Thermal Throttling, Quantum Electrodynamics (QED)*

MSEC-RS-2026-PHYSICS-006

Learning Challenges and Teaching Methods in Higher Education

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Abstract

This study examines the challenges that students face and the teaching methods used by their instructors. It is based on a survey completed by 35 first-year engineering students. Understanding how students learn and the difficulties they encounter is important for improving teaching in higher education. Many students struggle to grasp concepts due to different learning styles, teaching methods, and academic pressure. The survey looked at students' learning habits, the problems they encounter while studying, and their preferences for various teaching methods. The findings reveal that many students find fast-paced teaching difficult, lack of real-life examples frustrating, and unclear concepts confusing. The results also show that students learn better when teaching is interactive. They preferred methods like discussions, visual aids, and practical examples rather than traditional lectures. Many students mentioned that active participation and clear explanations help them learn more effectively. The study suggests that using more interactive and student-focused methods can improve understanding and make learning more meaningful. These insights can help educators create more effective and engaging learning environments in higher education.

Keywords: *understanding challenges, teaching methodology, real-life examples, conceptual clarity, active participation, engaging learning environments.*

MSEC-RS-2026-CHEM-001

Generation of Bioelectric Energy from Chlorophyll-Embedded Mesophyll Cells

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Abstract

The research investigates how plant tissues create bio- electric potential differences through the examination of mesophyll cells in chlorophyll-containing leaf tissues. We focus our research on obtaining electrons which plants release during their photosynthesis and respi- ration processes. The introduction of dissimilar metals establishes an electrochemical equilibrium between two electrodes which creates an electrochemical potential as leaf tissues and saturated soil absorb ions through intercalation. The system generates voltage which reacts to various environmental changes so we can evaluate multiple biological explanations. The results show that a micro-power system built from plants provides a practical solution for biologically based energy systems which supply power to ultra-low-energy electronic devices and sustainable green energy systems.

Keywords: *Bioelectric energy, Mesophyll cells, Chlorophyll, Bio electrochemical cell, Renewable energy*

MSEC-RS-2026-CHEM-002

Ai-Driven Autonomous Smart Composting System With Predictive Methane Mitigation and Circular Energy Optimization

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Abstract

The rapid increase in decentralized organic waste generation has intensified methane emissions from compost-based waste management systems. Conventional composting methods rely on manual monitoring and lack predictive environmental stabilization mechanisms. This work proposes an AI-driven autonomous smart composting system integrating real-time multi-parameter sensing, predictive methane modeling, closed-loop automated mitigation, and energy recovery estimation. Environmental parameters such as temperature, moisture, pH, methane concentration, ammonia concentration, and oxygen levels are continuously monitored using an IoT-based architecture. A regression-based predictive model estimates methane spike probability, while a novel Compost Environmental Stability Index (CESI) quantifies aerobic process stability. Upon detection of environmental instability, automated aeration and moisture regulation mechanisms are activated. The proposed system enhances compost stabilization efficiency, reduces greenhouse gas emissions, and supports scalable decentralized waste management deployment.

Keywords: *Smart Composting, Methane Prediction, IoT, Autonomous Control, Circular Economy, Sustainable Waste Management.*

MSEC-RS-2026-CHEM-003

Essential Oil Encapsulated Hybrid Electrospun Fibers – An Advanced Nano Packaging Technology

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Abstract

Here is a new tactical reach in bringing out the most efficient packaging material fabricated out of polymethylmethacrylate / polyurethane (PMMA /PU) polymeric blend incorporated with three different essential oils such as citronella essential oil (CEO), lavender essential oil (LEO) and peppermint essential oil (PEO). Thermograms of the blends proved the excellent thermal stability. As the stringy nature served as an important criteria for a packaging material, the fibrous morphology of the blend helped in enhancing the packing property of the material. The shelf life evaluation conducted on table grapes displayed the effect of EO in protecting and retaining the food item through its antibacterial nature. The observations revealed the competence of essential oil encapsulated blend membrane to be used as a nano packaging material for an extended shelf life.

Keywords: *essential oil; extended shelf life; hybrid fibers; nano-packaging; PU/PMMA*

MSEC-RS-2026-CHEM-004

Synthesis, Spectral, BVS, Crystallographic and Computational Studies of Homoleptic and Heteroleptic Precursor Complexes Involving Dithiocarbamates, Triphenylphosphine and Nickel(II): Synthesis of Nano Nickel Sulfide and Nickel Oxide from Precursors

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Abstract

[Ni(phqdtc)2] 1, [Ni(bpedtc)2] 2, [Ni(phqdtc)(NCS)(PPh3)] 3 and Ni(bpedtc)(NCS)PPh3] 4, (where phqdtc = perhydroisoquinoline dithiocarbamate, bpedtc = n-benzyl-2-phenethylamine dithiocarbamate) have been prepared and characterized by elemental analysis, electronic, IR and NMR (1H, 13C, and 31P) spectra. All the complexes are diamagnetic and planar. Single crystal X-ray was carried out for the complexes 3 and 4 and showed that nickel is in a distorted square planar arrangement with NiS2PN chromophore. Electronic spectra of the complexes show bands corresponding to dz2→dxy/dx2-y2 transitions. IR and 13C NMR studies indicate the mesomeric flow of π-electron density from the dithiocarbamate towards the nickel. The deshielding for the coordinated phosphorus signals in 31P NMR spectra for the complexes 3 and 4 compared with the free phosphine clearly manifests the drift of electron density from phosphorus towards the metal on complexation. Complexes 1 and 2 have been used as the single source precursors for the preparation of nickel sulfide and nickel oxide nanoparticles. The synthesized nickel sulfide and nickel oxide nano particles have been characterized by pXRD, SEM, EDAX and IR spectroscopy. Geometry optimization, geometrical parameters, molecular electrostatic potential maps (MEPs) and FMO analysis of the complex 3 and 4 was carried out by DFT method and the geometrical parameters are compared with the experimental X-ray diffraction data.

Keywords: *Nickel (II), dithiocarbamate, thioureide, thiocyanate-N, x-ray structure, nano sulfide, nano oxide*

MSEC-RS-2026-ENG-001

Women Characters in R. K. Narayan's Novels

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Abstract

Women characters in R.K. Narayan's novels occupy a central yet understated space that reflects the socio-cultural realities of traditional South Indian life. Narayan portrays women not as passive figures but as individuals negotiating personal desires within the constraints of family, community, and custom. Characters such as Savitri in *The Dark Room*, Daisy in *The Painter of Signs*, and Rosie in *The Guide* illustrate a spectrum of female experiences—from domestic oppression and emotional restraint to self-assertion, professional independence, and artistic freedom. Through subtle narrative techniques and gentle irony, Narayan reveals the tensions between tradition and modernity, highlighting how women's identities evolve as they challenge patriarchal expectations. His female protagonists often emerge as transformative agents who influence not only their own destinies but also the moral growth of the male characters. Overall, Narayan's depiction of women is marked by empathy, realism, and a quiet critique of societal norms, making his work an important commentary on gender roles in early 20th-century Indian society.

Keywords: *socio-cultural realities, spectrum of female experiences, domestic oppression and emotional restraint to self-assertion, professional independence, and artistic freedom.*

MSEC-RS-2026-ENG-002

Technological Transgression and Ethical Limits: A Transhumanist Reading of *Frankenstein*

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Abstract

Mary Shelley's *Frankenstein*, or *The Modern Prometheus*, is among the most notable works of English literature. The novel has been interpreted using a variety of critical frameworks since it was published in 1818, including feminism, materialism, Marxism, and psychological critique. Its significance within the context of transhumanism, however, has received comparatively little attention. There are new ways to understand Shelley's work in contemporary contexts due to transhumanism, which is roughly defined as the fusion of biotechnology, information technology, nanotechnology, and cognitive research to improve human capabilities. By comparing Victor Frankenstein's creation to contemporary advancements in artificial intelligence (Al), this article aims to examine the ethical implications and constraints of scientific progress through a transhumanist viewpoint. The Monster raises issues with human ambition, accountability, and the unforeseen repercussions of invention. It can be viewed as an early example of technologically enhanced life. The study critically investigates if Al-driven technologies, like the Monster, actually benefit humans or if they could endanger ecological equilibrium and the natural lifecycle. The study also examines the conflict between ethical responsibility and technical advancement, highlighting the potentially disastrous consequences of unbridled scientific endeavors. In the end, this study emphasizes how important it is to set moral limits for scientific research in order to ensure that innovation benefits both people and the environment.

Keywords: *Frankenstein, Artificial Intelligence, Transhumanism, Dark science, Ecology, Ethics, Technology, Responsibility*

MSEC-RS-2026-ENG-003

Human Vs. Machine Writing: A Literary Analysis of Ai-Generated Texts

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Abstract

The development of artificial intelligence has brought major changes to the way texts are written and understood. This paper, Human vs. Machine Writing: A Literary Analysis of AI-Generated Texts, explores the key differences between human writing and AI-generated writing while examining their limitations and possibilities, with the aim of understanding whether machine-generated texts can be considered literary and how they compare with human creativity. The study uses a comparative method by analyzing selected human-written and AI-generated texts, focusing on key literary aspects such as language use, creativity, emotional expression, coherence, and depth of meaning, along with reader response. The analysis shows that human writing is shaped by personal experience, emotions, and cultural background, which gives it deeper meaning, originality, and emotional richness, allowing writers to express complex thoughts, ambiguity, and unique perspectives. In contrast, AI-generated writing is based on patterns and data; although it is grammatically correct and well-structured, it often lacks genuine emotion, personal voice, and true originality, and may appear repetitive or general. At the same time, AI writing has notable strengths: it is fast, efficient, and capable of generating large amounts of text, making it useful for idea generation, drafting, and language support in academic and professional contexts. The paper also highlights that human writing can be time-consuming and influenced by bias or limited knowledge, while AI lacks consciousness, intention, and real-life experience. Despite these limitations, AI opens new possibilities for collaboration, supporting rather than replacing human creativity. The study concludes that human and machine writing should be seen not simply as opposites but as complementary forms with their own strengths, calling for a broader understanding of creativity in the digital age.

Keywords: *AI, Human Writing, Machine-Generated Texts, Literary Analysis, Creativity*

MSEC-RS-2026-ENG-004

The Post-Modern Vampire in Anne Rice's Vampire Chronicles

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Abstract

Anne Rice's postmodern book The Vampire Chronicles breaks the rules of centuries-old vampire stories by exploring what it means to be human in a world that is unsure and broken. This article looks at postmodern problems like identity, morality, power, desire, and cultural critique through the lens of Lestat de Lioncourt, Claudia, Louis de Pointe du Lac, and Rice's vampires. These traits are carefully looked at. The vampires that Rice writes about break the rules by having unclear personalities and moral problems. Start the process of losing traditions. Because there are no absolutes in their world, Louis and Lestat have psychological crises where they think about where they fit in the universe. Louis's moral worries are about social morality and basic principles, while Lestat is the poster child for postmodernism's disobedience and existential pain. By making connections between her vampire figures and people, Rice questions cultural norms and criticisms. In its exploration of desire, power, and decadence, the story shows the major moral issues that plague the vampire lifestyle, despite its appeal. Using the supernatural as a metaphor, Rice makes people think about what it means to be human, to be strong, and to enjoy life. This is done by looking at modern culture. By mixing fact and fiction, intertextuality and metafictional techniques make the story stronger and encourage the reader to think critically about it. Postmodernists are interested in language and how it is represented. Finally, Anne Rice's The Vampire Chronicles are an interesting look at postmodern vampire writing. Through the show's complex intertextuality and thought-provoking social commentary, viewers are forced to deal with difficult moral dilemmas, find meaning in a world where fiction and reality are constantly changing, and watch as multiple characters grow.

Keywords: *Postmodern, Vampire, Identity, Cultural critique, Existential angst*

Proceedings of Fourth Research Summit 2026
Explore, Evaluate & Extend (RSEEE'26)

MSEC-RS-2026-ENG-005

Defense Mechanisms in Colleen Mccullough's *The Thorn Birds*: A Psychoanalytic Study

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Abstract

This paper attempts a psychoanalytic study of the major characters in *The Thorn Birds* by Colleen McCullough, an Australian writer. The study focuses on the use of ego defense mechanisms by the characters. Under the light of psychoanalytic study, especially, Sigmund Freud and Anne Freud's Defense Mechanism, the study examines how repression, denial, displacement, projection, and sublimation operate in the lives of Meggie Cleary, Father Ralph de Bricassart, and Fiona Cleary. The analysis reveals that the emotional conflicts of these characters arise from continuous inner struggle. Meggie's repeated emotional imbalances lead her to rely heavily on repression, denial, and sublimation. On the other hand, Ralph's ambition and religious identity result in denial and repression of natural human desires. Fiona's emotional detachment is rooted in guilt and expressed through repression and projection. The study shows that defense mechanisms are not signs of weakness but natural psychological responses to suffering and social pressure. Through this psychoanalytic reading, the paper highlights how McCullough realistically portrays psychological conflict and how the characters cope with their conflicts with ego's defense mechanisms.

Keywords: *Psychoanalysis, Defense Mechanisms, Repression, Denial, Sublimation, The Thorn Birds*

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